



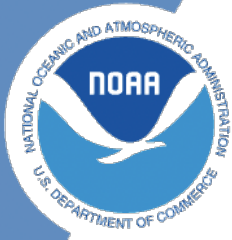
NOAA Technical Memorandum NMFS-XXX-##

NOAA quarto book

Mandy Karnauskas

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U.S. DEPARTMENT OF COMMERCE
National Oceanic and Atmospheric
Administration
National Marine Fisheries Service
Northwest Fisheries Science Center



**NOAA
FISHERIES**

NOAA quarto book

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1. NOAA Fisheries, Southeast Fisheries Science Center

Table of contents

1	Introduction	1
1.1	Details	1
2	NOAA Fisheries recreational landings data	2
2.1	Data access and upload	2
2.2	Separate Gulf and Atlantic recreational landings	5
2.3	Partition catches by area and sector	9
2.4	Looking at extent of discarding	17
2.5	Split the catches by season	18
3	Analysis of international landings estimates from Sea Around Us database	40
3.1	Data access and upload	40
3.2	Read in the high seas landings	40
3.3	Summarize the high seas landings	45
3.4	Read in the landings within exclusive economic zones (EEZs)	48
3.5	Separating the EEZs by area	50
3.6	Summarize catch by regional EEZs	57
3.7	Merge high seas with catch with exclusive economic zones	59
3.8	Plot Western Atlantic catches	62
3.9	Look at Venezuela catches in detail	65
3.10	References	71
4	NOAA Fisheries landings data	72
4.1	Data access and upload	72
4.2	Summarize the landings	73
4.3	Plot the summary	75
4.4	Updated Gulf/Atlantic parsing of data	76
4.5	Plots with updated data	79
5	SST	81

Table of contents

6	Analysis of ocean temperature trends	82
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List of Figures

List of Tables

1 Introduction

Here we will be displaying various data summaries and analyses related to dolphin management in the U.S. South Atlantic.

1.1 Details

The documentation of this data exploration was developed in R, using R Studio and Quarto.

2 NOAA Fisheries recreational landings data

Here we will partition recreational landings data from NOAA Fisheries' Marine Recreational Information Program (MRIP), a standardized, peer-reviewed suite of recreational fishing surveys used to produce catch and effort estimates for quota monitoring and stock assessment in the United States. MRIP is a standardized survey suite that employs a complex statistical sampling design stratified across time, regions of the coast, and different fishing modes, which allows the intercept data to be scaled up by effort estimates to calculate total landings.

2.1 Data access and upload

Data are downloaded directly from the Miami server using the files provided to SERO for ACL monitoring. The current location is: M://SFD/SECM-SFD/ACL/2025_Aug19_-MRIP_FES/. The landings are in FES MRIP and the units are in pounds whole weight.

```
# clear workspace
rm(list = ls())

# read in data file
load("data/mrip_fes_rec81_25wv2_23June25.RData") # most recent file on SFD server
head(dat)
```

	REC_ACL_MRIP_FES_ID	YEAR	MONTH	WAVE	SUB_REG	NEW_ST	NEW_STA	NEW_MODE	NEW_MODEN
1	73407425	1989	NA	4	6	9	NC	4	Priv
2	73407426	1989	NA	4	6	9	NC	4	Priv
3	73407427	1989	NA	4	6	9	NC	4	Priv
4	73407428	1989	NA	4	6	9	NC	4	Priv
5	73407429	1989	NA	4	6	9	NC	4	Priv
6	73407430	1989	NA	4	6	9	NC	4	Priv

	AREA_X	NEW_AREA	NEW_AREAN	FL_REG	NC_REG	DS	NEW_SCI
1	5	5	Inshore	NA	S	MRIP Centropristis	striata
2	5	5	Inshore	NA	N	MRIP Centropristis	striata

2 NOAA Fisheries recreational landings data

3	2	2	Ocean>3mi	NA	N MRIP	Centropristis striata
4	2	2	Ocean>3mi	NA	S MRIP	Centropristis striata
5	1	1	Ocean<=3mi	NA	N MRIP	Centropristis striata
6	1	1	Ocean<=3mi	NA	S MRIP	Centropristis striata
	NEW_COM	SP_CODE	SPECIES_ITIS	SPECIES	AB1	B2
1	black sea bass	8835020301	167687		34181.4033	121019.1368
2	black sea bass	8835020301	167687		795.8612	4445.0527
3	black sea bass	8835020301	167687		549.5390	2919.4840
4	black sea bass	8835020301	167687		71823.2390	24548.2535
5	black sea bass	8835020301	167687		415.7642	87.5928
6	black sea bass	8835020301	167687		48728.5632	11859.6428
	A	B1	LBSEST_SECWWT	LBSEST_SECSOURCE	SAMPLE_SIZE_USED	
1	31761.4755	2419.9278	11457.8425	srysmwa	41	
2	584.4155	211.4457	266.7782	srysmwa	41	
3	549.5390	0.0000	484.4319	srysmwa	119	
4	33343.1844	38480.0546	63313.9276	srysmwa	119	
5	415.7642	0.0000	294.0623	srysmwa	71	
6	40585.0152	8143.5480	34464.8156	srysmwa	71	
	AVG_LEN	AVG_WGT	SRYSMWA_AVG_LEN	SRYSMWA_AVG_WGT	SRYSMWA_SAMPLE_SIZE	
1	203.4572	0.3352069	203.4572	0.3352069	41	
2	203.4572	0.3352069	203.4572	0.3352069	41	
3	296.2590	0.8815243	296.2590	0.8815243	119	
4	296.2590	0.8815243	296.2590	0.8815243	119	
5	261.9949	0.7072816	261.9949	0.7072816	71	
6	261.9949	0.7072816	261.9949	0.7072816	71	
	SRYSMWA_FLAGGED	SRYSMW_AVG_LEN	SRYSMW_AVG_WGT	SRYSMW_SAMPLE_SIZE		
1		NA	269.2563	0.7310037	231	
2		NA	269.2563	0.7310037	231	
3		NA	269.2563	0.7310037	231	
4		NA	269.2563	0.7310037	231	
5		NA	269.2563	0.7310037	231	
6		NA	269.2563	0.7310037	231	
	SRYSMW_FLAGGED	SRYSMW_AVG_LEN	SRYSMW_AVG_WGT	SRYSMW_SAMPLE_SIZE	SRYSMW_FLAGGED	
1		NA	259.7694	0.6948228	556	NA
2		NA	259.7694	0.6948228	556	NA
3		NA	259.7694	0.6948228	556	NA
4		NA	259.7694	0.6948228	556	NA
5		NA	259.7694	0.6948228	556	NA
6		NA	259.7694	0.6948228	556	NA
	SRYS_AVG_LEN	SRYS_AVG_WGT	SRYS_SAMPLE_SIZE	SRYS_FLAGGED	SRYS_AVG_LEN	
1	270.955	0.8099641	982	NA	269.1276	

2 NOAA Fisheries recreational landings data

2	270.955	0.8099641		982	NA	269.1276		
3	270.955	0.8099641		982	NA	269.1276		
4	270.955	0.8099641		982	NA	269.1276		
5	270.955	0.8099641		982	NA	269.1276		
6	270.955	0.8099641		982	NA	269.1276		
	SRY_AVG_WGT	SRY_SAMPLE_SIZE	SRY_FLAGGED	SR_AVG_LEN	SR_AVG_WGT	SR_SAMPLE_SIZE		
1	0.8085211	1703	NA	303.6871	1.050403	46055		
2	0.8085211	1703	NA	303.6871	1.050403	46055		
3	0.8085211	1703	NA	303.6871	1.050403	46055		
4	0.8085211	1703	NA	303.6871	1.050403	46055		
5	0.8085211	1703	NA	303.6871	1.050403	46055		
6	0.8085211	1703	NA	303.6871	1.050403	46055		
	SR_FLAGGED	S_AVG_LEN	S_AVG_WGT	S_SAMPLE_SIZE	S_FLAGGED	VAR_B2	VAR_AB1	
1	NA	344.4373	1.364082	305089	NA	4072492794	310099094	
2	NA	344.4373	1.364082	305089	NA	2866289	258947	
3	NA	344.4373	1.364082	305089	NA	0	0	
4	NA	344.4373	1.364082	305089	NA	105731413	534702898	
5	NA	344.4373	1.364082	305089	NA	0	0	
6	NA	344.4373	1.364082	305089	NA	22265768	286196169	
	SA_LABEL	GOM_LABEL	REC_ACL	MIGRA_GRP	AGG_MODEN	JURISDICTION	PS_REG	
1	Snapper	Grouper	1		Private	State	6	
2	Snapper	Grouper	1		Private	State	5	
3	Snapper	Grouper	1		Private	Federal	5	
4	Snapper	Grouper	1		Private	Federal	6	
5	Snapper	Grouper	1		Private	State	5	
6	Snapper	Grouper	1		Private	State	6	
	COUNCILREG	GFMC_FMP	SAFMC_FMU	LBSEST_SECGWT	AGGR_AREA	AREA	FIRST_MONTH	
1	6		Y	9710.0360		NA	NA	
2	5		Y	226.0832		NA	NA	
3	5		Y	410.5355		NA	NA	
4	6		Y	53655.8708		NA	NA	
5	5		Y	249.2054		NA	NA	
6	6		Y	29207.4709		NA	NA	
	LAST_MONTH	ALT_FLAG	CHTS_CL	CHTS_H	CHTS_RL	CHTS_WAB1C	CHTS_WAB1H	MODE_FX
1	NA	0	0	0	0	0	0	7
2	NA	0	0	0	0	0	0	7
3	NA	0	0	0	0	0	0	7
4	NA	0	0	0	0	0	0	7
5	NA	0	0	0	0	0	0	7
6	NA	0	0	0	0	0	0	7
	NEW_SEASN	SEASON						

2 NOAA Fisheries recreational landings data

```
1      NA
2      NA
3      NA
4      NA
5      NA
6      NA
```

```
#apply(dat[2:18], 2, table, useNA = "always")
```

```
# confirm nomenclature for dolphin and subset by species -----
table(dat$NEW_COM[grepl("dolphin", dat$NEW_COM)])
```

```
dolphin
9377
```

```
table(dat$NEW_SCI[grepl("dolphin", dat$NEW_COM)])
```

```
Coryphaena hippurus
9377
```

```
d <- dat[which(dat$NEW_COM == "dolphin"), ]
#apply(d[2:18], 2, table, useNA = "always")
d <- d[which(d$YEAR < 2025), ] # take out 2025 because incomplete year
```

2.2 Separate Gulf and Atlantic recreational landings

Now we filter the data set to separate out the Gulf versus Atlantic landings.

```
# codes for WFL regions: 1 (NW FL Panhandle- Escambia to Dixie co.), 2 (SW FL Peninsula- Le
table(d$FL_REG, useNA = "always")
```

```
1    2    3    4    5 <NA>
387 110 568 1022 312 6943
```

2 NOAA Fisheries recreational landings data

```
# subregional codes: 6 is Atlantic and 7 is Gulf
table(d$FL_REG, d$SUB_REG, useNA = "always")
```

	4	5	6	7	<NA>
1	0	0	0	387	0
2	0	0	0	110	0
3	0	0	0	568	0
4	0	0	1022	0	0
5	0	0	312	0	0
<NA>	96	585	3836	2426	0

```
unique(d$NEW_STA)
```

```
[1] "NC"      "FLE"      "FLW"      "SC"      "GA"      "MS"      "AL"      "LA"
[9] "TX"      "FLE/GA"   "RI"       "VA"      "DE"      "MD"      "NJ"      "CT"
[17] "MA"      "NY"       "LA/MS"    "AL/FLW"
```

```
lis <- c("TX", "LA", "LA/MS", "MS", "AL", "AL/FLW") # filter out the Gulf states
d1 <- d[-which(d$NEW_STA %in% lis), ]
unique(d1$NEW_STA)
```

```
[1] "NC"      "FLE"      "FLW"      "SC"      "GA"      "FLE/GA"   "RI"      "VA"
[9] "DE"      "MD"       "NJ"      "CT"      "MA"      "NY"
```

```
table(d1$FL_REG, d1$NEW_STA, useNA = "always")
```

	CT	DE	FLE	FLE/GA	FLW	GA	MA	MD	NC	NJ	NY	RI	SC	VA
1	0	0	0	0	387	0	0	0	0	0	0	0	0	0
2	0	0	0	0	110	0	0	0	0	0	0	0	0	0
3	0	0	0	0	568	0	0	0	0	0	0	0	0	0
4	0	0	1022	0	0	0	0	0	0	0	0	0	0	0
5	0	0	312	0	0	0	0	0	0	0	0	0	0	0
<NA>	14	114	0	1794	174	54	20	125	1344	108	73	62	644	165

<NA>

2 NOAA Fisheries recreational landings data

```
1      0
2      0
3      0
4      0
5      0
<NA>   0
```

```
dim(d1)
```

```
[1] 7090   83
```

```
d1 <- d1[-which(d1$FL_REG == 1 | d1$FL_REG == 2), ] # filter out FL regions 1 and 2 but r
dim(d1)
```

```
[1] 6593   83
```

```
d1 <- d1[-which(is.na(d1$FL_REG) & d1$NEW_STA == "FLW"), ] # filter out FLW with unknown F
dim(d1)
```

```
[1] 6419   83
```

```
table(d1$FL_REG, d1$NEW_STA, useNA = "always") # retained are all Atlantic states plus reg
```

	CT	DE	FLE	FLE/GA	FLW	GA	MA	MD	NC	NJ	NY	RI	SC	VA
3	0	0	0	0	568	0	0	0	0	0	0	0	0	0
4	0	0	1022	0	0	0	0	0	0	0	0	0	0	0
5	0	0	312	0	0	0	0	0	0	0	0	0	0	0
<NA>	14	114	0	1794	0	54	20	125	1344	108	73	62	644	165

```
<NA>
3      0
4      0
5      0
<NA>   0
```

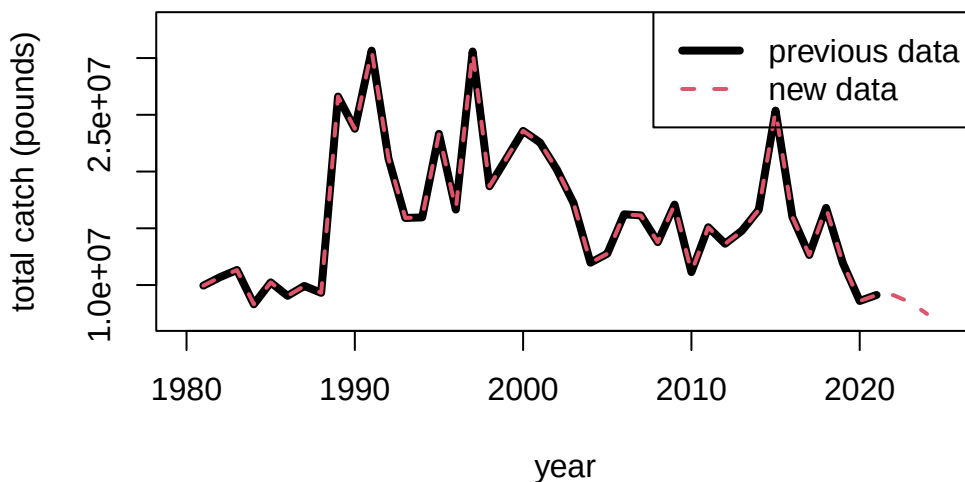
2 NOAA Fisheries recreational landings data

Now we can summarize the total dolphin landings in weight for the U.S. Atlantic, and compare to previous values sent for quota monitoring. We see that the values are almost identical.

```
# summarize by year and compare -----  
  
tot1 <- tapply(d1$LBSEST_SECWWT, d1$YEAR, sum, na.rm = T)  
head(tot1)
```

```
      1981      1982      1983      1984      1985      1986  
9952644 10701659 11325575  8303073 10236339  9047438
```

```
old <- read.csv("data/RecreationalDolphinLandings_SAandGOM_MLarkin.csv", skip = 6)  
plot(old$Year, old$ATL, type = "l", xlab = "year", ylab = "total catch (pounds)", lwd = 4,  
      xlim = c(1980, 2025), ylim = c(7*10^6, 3.3*10^7))  
lines(as.numeric(names(tot1)), tot1, lwd = 2, col = 2, lty = 2)  
legend("topright", c("previous data", "new data"), col = c(1, 2), lwd = c(4, 2), lty = c(1,
```



2.3 Partition catches by area and sector

Now we will partition out the total catches according to the dolphin MSE areas. The areas are as follows:

- FLK: Florida Keys to Indian River County County (FL)
- NCFL: Brevard (FL) to Southern NC (South of Hatteras)
- NNC: Northern NC (North of Hatteras to NC/VA border)
- VBM: Virginia to Maine

```
# separate out 4 regions -----

d1$region <- "VBM"

# select for FLK
# FL_REG codes: 4: SE FL- Miami-Dade to Indian River County.; 5: NE FL- Brevard to Nassau C
table(d1$FL_REG, d1$NEW_STA, useNA = "always")
```

	CT	DE	FLE	FLE/GA	FLW	GA	MA	MD	NC	NJ	NY	RI	SC	VA
3	0	0	0	0	568	0	0	0	0	0	0	0	0	0
4	0	0	1022	0	0	0	0	0	0	0	0	0	0	0
5	0	0	312	0	0	0	0	0	0	0	0	0	0	0
<NA>	14	114	0	1794	0	54	20	125	1344	108	73	62	644	165

	<NA>
3	0
4	0
5	0
<NA>	0

```
d1$region[which(d1$FL_REG == 3 | d1$FL_REG == 4)] <- "FLK"

# select for NCFL
d1$region[which(d1$FL_REG == 5)] <- "NCFL"
lis <- c("SC", "GA", "FLE/GA") # NCFL states
d1$region[which(d1$NEW_STA %in% lis)] <- "NCFL"
# NC_REG codes: N: North of Cape Hatteras; S: South of Cape Hatteras
table(d1$NC_REG, d1$NEW_STA, useNA = "always")
```

2 NOAA Fisheries recreational landings data

	CT	DE	FLE	FLE/GA	FLW	GA	MA	MD	NC	NJ	NY	RI	SC	VA
	14	114	1334	1794	568	54	20	125	586	108	73	62	644	165
N	0	0	0	0	0	0	0	0	420	0	0	0	0	0
S	0	0	0	0	0	0	0	0	338	0	0	0	0	0
<NA>	0	0	0	0	0	0	0	0	0	0	0	0	0	0

```

<NA>
0
N      0
S      0
<NA>  0

```

```

d1$region[which(d1$NC == "S")] <- "NCFL"

# select for NNC
d1$region[which(d1$NC == "N")] <- "NNC"

table(d1$FL_REG, d1$NEW_STA, d1$region)

```

```
, , = FLK
```

	CT	DE	FLE	FLE/GA	FLW	GA	MA	MD	NC	NJ	NY	RI	SC	VA
3	0	0	0	0	568	0	0	0	0	0	0	0	0	0
4	0	0	1022	0	0	0	0	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0	0	0	0	0	0	0	0

```
, , = NCFL
```

	CT	DE	FLE	FLE/GA	FLW	GA	MA	MD	NC	NJ	NY	RI	SC	VA
3	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5	0	0	312	0	0	0	0	0	0	0	0	0	0	0

```
, , = NNC
```

```

CT  DE  FLE  FLE/GA  FLW  GA  MA  MD  NC  NJ  NY  RI  SC  VA

```


2 NOAA Fisheries recreational landings data

3	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0	0	0	0	0	0	0	0

, , = VBM

	CT	DE	FLE	FLE/GA	FLW	GA	MA	MD	NC	NJ	NY	RI	SC	VA
3	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0	0	0	0	0	0	0	0

```
table(d1$NC_REG, d1$NEW_STA, d1$region) # check classification
```

, , = FLK

	CT	DE	FLE	FLE/GA	FLW	GA	MA	MD	NC	NJ	NY	RI	SC	VA
	0	0	1022	0	568	0	0	0	0	0	0	0	0	0
N	0	0	0	0	0	0	0	0	0	0	0	0	0	0
S	0	0	0	0	0	0	0	0	0	0	0	0	0	0

, , = NCFL

	CT	DE	FLE	FLE/GA	FLW	GA	MA	MD	NC	NJ	NY	RI	SC	VA
	0	0	312	1794	0	54	0	0	0	0	0	0	644	0
N	0	0	0	0	0	0	0	0	0	0	0	0	0	0
S	0	0	0	0	0	0	0	0	338	0	0	0	0	0

, , = NNC

	CT	DE	FLE	FLE/GA	FLW	GA	MA	MD	NC	NJ	NY	RI	SC	VA
	0	0	0	0	0	0	0	0	0	0	0	0	0	0
N	0	0	0	0	0	0	0	0	420	0	0	0	0	0
S	0	0	0	0	0	0	0	0	0	0	0	0	0	0

, , = VBM

2 NOAA Fisheries recreational landings data

	CT	DE	FLE	FLE/GA	FLW	GA	MA	MD	NC	NJ	NY	RI	SC	VA
	14	114	0	0	0	0	20	125	586	108	73	62	0	165
N	0	0	0	0	0	0	0	0	0	0	0	0	0	0
S	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Now we will separate out the catches by recreational sector and plot the totals by region and sector.

```
# separate out by for hire vs. private rec -----
d1$sector <- "ForHire"

d1$sector[which(d1$NEW_MODEN == "Shore")] <- "Rec"
d1$sector[which(d1$NEW_MODEN == "Priv")] <- "Rec"

table(d1$sector, d1$NEW_MODEN, useNA = "always") # check classification
```

	Cbt	Cbt/Hbt	Hbt	Priv	Shore	<NA>
ForHire	1616	119	2912	0	0	0
Rec	0	0	0	1750	22	0
<NA>	0	0	0	0	0	0

```
tot <- tapply(d1$LBSEST_SECWWT, list(d1$YEAR, d1$region, d1$sector), sum, na.rm = T)

head(tot)
```

, , ForHire

	FLK	NCFL	NNC	VBM
1981	354639.5	74702.50	256755.24	1400.247
1982	347458.2	92271.42	27578.44	2749.224
1983	1020005.8	41784.04	88690.53	5379.177
1984	1021086.8	74658.81	NA	4203.048
1985	717523.0	172989.65	58514.13	8880.857
1986	1562354.0	63555.89	817192.26	61658.193

, , Rec

2 NOAA Fisheries recreational landings data

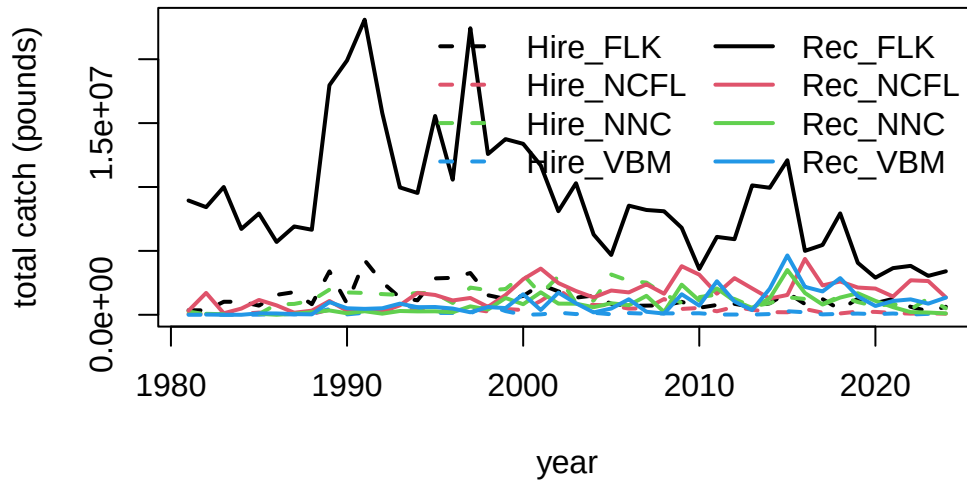
	FLK	NCFL	NNC	VBM
1981	8942641	322505.7	NA	NA
1982	8422673	1711834.7	78367.161	18727.290
1983	10000662	119370.4	44226.032	5456.237
1984	6717044	486079.7	NA	0.000
1985	7925301	1162708.4	77477.096	112944.274
1986	5699311	736872.0	2549.586	103945.134

```
fin <- data.frame(cbind(tot[, , 1], tot[, , 2]))
names(fin) <- c("Hire_FLK", "Hire_NCFL", "Hire_NNC", "Hire_VBM", "Rec_FLK", "Rec_NCFL", "Rec_NNC")
head(fin)
```

	Hire_FLK	Hire_NCFL	Hire_NNC	Hire_VBM	Rec_FLK	Rec_NCFL	Rec_NNC
1981	354639.5	74702.50	256755.24	1400.247	8942641	322505.7	NA
1982	347458.2	92271.42	27578.44	2749.224	8422673	1711834.7	78367.161
1983	1020005.8	41784.04	88690.53	5379.177	10000662	119370.4	44226.032
1984	1021086.8	74658.81	NA	4203.048	6717044	486079.7	NA
1985	717523.0	172989.65	58514.13	8880.857	7925301	1162708.4	77477.096
1986	1562354.0	63555.89	817192.26	61658.193	5699311	736872.0	2549.586
	Rec_VBM						
1981	NA						
1982	18727.290						
1983	5456.237						
1984	0.000						
1985	112944.274						
1986	103945.134						

```
matplot(rownames(fin), fin, lty = c(rep(2, 4), rep(1, 4)), col = rep(1:4, 2), type = "l", las = 1,
        xlab = "year", ylab = "total catch (pounds)")
legend("topright", colnames(fin), lty = c(rep(2, 4), rep(1, 4)), col = rep(1:4, 2), lwd = 2)
```

2 NOAA Fisheries recreational landings data



```
# check that totals match
cor(rowSums(fin, na.rm = T), tot1)
```

```
[1] 1
```

```
mean(rowSums(fin, na.rm = T) - tot1) # very tiny differences
```

```
[1] 4.233284e-11
```

Here are the catches (in pounds) for years 1981 to 2024.

```
fin
```

	Hire_FLK	Hire_NCFL	Hire_NNC	Hire_VBM	Rec_FLK	Rec_NCFL	Rec_NNC
1981	354639.5	74702.50	256755.24	1400.247	8942641	322505.7	NA
1982	347458.2	92271.42	27578.44	2749.224	8422673	1711834.7	78367.161
1983	1020005.8	41784.04	88690.53	5379.177	10000662	119370.4	44226.032
1984	1021086.8	74658.81	NA	4203.048	6717044	486079.7	NA
1985	717523.0	172989.65	58514.13	8880.857	7925301	1162708.4	77477.096

2 NOAA Fisheries recreational landings data

1986	1562354.0	63555.89	817192.26	61658.193	5699311	736872.0	2549.586
1987	1774558.7	61243.58	848994.25	24967.877	6913585	161627.1	79540.365
1988	826480.6	62392.87	1114627.05	94676.212	6655923	321023.1	202768.950
1989	3398703.5	405140.01	1970776.23	451477.768	17965047	1079951.5	339371.567
1990	868572.6	277783.17	1749965.55	31079.442	19900865	312333.3	118539.960
1991	4268773.7	225552.64	1706751.50	159172.855	23093044	457667.1	277389.646
1992	2548407.4	189514.34	1619685.06	161149.703	15788853	245395.8	89160.860
1993	1343093.6	391254.33	1555850.09	741955.369	9967356	742741.8	296399.199
1994	1131831.7	437642.72	1713981.98	573551.716	9532763	1708942.1	264907.193
1995	2848691.7	667475.30	1673435.48	145798.587	15558489	1550136.0	271442.073
1996	2892436.3	309604.42	923128.81	141976.431	10574454	1121937.8	196579.537
1997	3271271.6	466891.67	2129471.46	110115.485	22423256	1316502.7	660257.843
1998	1532460.0	252620.73	1921834.08	779364.855	12578694	619824.1	412304.292
1999	1238462.0	457880.10	2014752.29	288625.130	13745135	1548217.4	1314805.374
2000	1471481.3	369553.33	3139555.98	3016.589	13379647	2784424.2	826980.337
2001	2361751.6	1090421.00	1638707.54	34335.241	11690170	3614370.3	1760721.144
2002	1828918.7	1993778.99	3005172.92	168406.810	8105262	2483946.6	871930.660
2003	1342339.1	958997.12	928894.01	55690.419	10294918	1860435.4	868200.254
2004	1509660.5	771820.65	1116373.11	155510.796	6281794	1354303.7	593257.716
2005	858090.6	734896.20	3153480.89	38020.553	4681210	1898945.9	904360.210
2006	700089.5	483337.12	2656240.22	136715.477	8540432	1751853.5	749541.413
2007	718651.9	506776.30	2518430.49	84908.300	8200488	2383417.1	1483630.590
2008	736213.2	1235299.71	1620865.87	113592.877	8100550	1648662.2	223565.185
2009	1016583.6	458562.82	928419.86	106262.915	6809447	3805127.7	2359177.817
2010	552080.9	534671.06	1384434.93	120050.410	3579591	3139199.6	1110479.730
2011	768597.7	275546.65	1577019.96	17844.729	6097050	1651096.4	2063315.642
2012	856261.4	615430.34	1236410.76	21887.318	5916144	2867599.1	1082607.063
2013	614150.1	394570.11	640465.60	16064.763	10124799	2064256.6	540676.961
2014	897508.4	210198.76	916695.96	56760.758	9940020	1272477.3	1289704.416
2015	1729825.9	196326.97	1374908.91	277867.986	12093928	1548428.5	3505194.743
2016	841662.5	454053.51	1236565.76	195837.854	5002273	4372622.6	1704661.669
2017	1227827.8	161188.31	835080.86	31185.061	5464893	2297930.0	801324.151
2018	401703.0	105328.49	1465087.58	72211.785	7935729	2606091.5	1347724.203
2019	1261827.1	250624.40	998102.45	93718.699	4064867	2132483.1	1687239.863
2020	910431.4	225493.79	667501.79	51621.152	2906913	2052222.0	1089245.171
2021	1244842.2	146893.15	871760.94	100997.979	3652596	1428508.3	596672.480
2022	627020.6	90504.71	427953.34	18514.133	3828113	2701619.8	211394.537
2023	274532.5	118528.80	1281254.55	72728.020	3045801	2639445.1	175189.224
2024	629327.2	85572.19	506558.07	18752.239	3404227	1358519.1	106307.278
Rec_VBM							
1981	NA						

2 NOAA Fisheries recreational landings data

1982	18727.290
1983	5456.237
1984	0.000
1985	112944.274
1986	103945.134
1987	52139.772
1988	33566.163
1989	996945.390
1990	509424.981
1991	464786.733
1992	509297.261
1993	871948.601
1994	594467.414
1995	609302.619
1996	486106.187
1997	197871.963
1998	606769.298
1999	523623.688
2000	1608478.920
2001	372623.273
2002	1732356.036
2003	904780.558
2004	186538.567
2005	489236.451
2006	1214497.040
2007	236769.123
2008	96818.461
2009	1607919.004
2010	717410.858
2011	2626410.052
2012	1041694.282
2013	364492.286
2014	2040329.550
2015	4645796.249
2016	2186126.672
2017	1826822.849
2018	2871124.410
2019	1440420.940
2020	689099.471
2021	1083268.811
2022	1206278.324

2 NOAA Fisheries recreational landings data

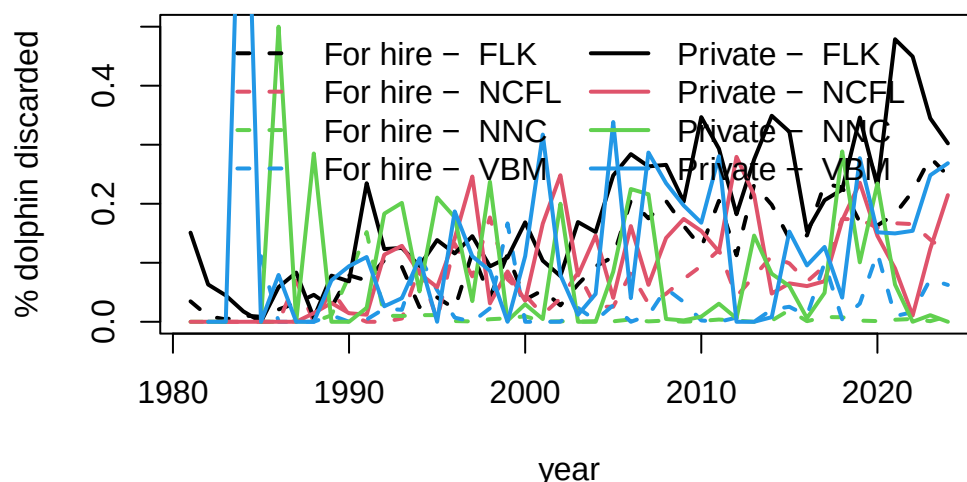
2023 872490.886
2024 1346606.842

2.4 Looking at extent of discarding

Discard estimates (known as B2 estimates) are provided in numbers by the MRIP survey but are not available in the Southeast Region Headboat Survey. We can look at the overall numbers of dolphinfish caught versus reported released from the MRIP survey to get an understanding of the extent of discarding.

```
d1$perdis <- d1$B2 / (d1$AB1 + d1$B2) # percent discarded is B2 (fish released alive) / to  
  
# calculate mean discard rate by year, region, sector -----  
dis <- tapply(d1$perdis, list(d1$YEAR, d1$region, d1$sector), mean, na.rm = T)  
disr <- tapply(d1$perdis, list(d1$region, d1$sector), mean, na.rm = T)  
  
matplot(rownames(dis), cbind(dis[, , 1], dis[, , 2]), lty = c(rep(2, 4), rep(1, 4)), col =  
        type = "l", lwd = 2, xlab = "year", ylab = "% dolphin discarded", ylim = c(0, 0.5))  
legend("top", c(paste("For hire - ", colnames(dis)), paste("Private - ", colnames(dis))),  
      lty = c(rep(2, 4), rep(1, 4)), col = rep(1:4, 2), lwd = 2, ncol = 2, bty = "n")
```

2 NOAA Fisheries recreational landings data



```
round(disr * 100, 2) # discarding rates by region and sector
```

	ForHire	Rec
FLK	10.79	18.76
NCFL	9.44	11.23
NNC	0.94	9.62
VBM	2.94	12.69

According to the MRIP survey and the reported numbers, discarding generally increases over time, excluding some highly anomalous estimates from the early years which are driven by low sample sizes. Discarding is generally higher in the private sector compared to the for-hire sector, and is highest in the Florida Keys region.

2.5 Split the catches by season

The dolphin MSE operates on a seasonal time step and thus the catches and discards must be parsed apart by individual season and year. This is not straightforward because MRIP estimates are reported in two-month waves (Jan - Feb, Mar - Apr, May -

2 NOAA Fisheries recreational landings data

Jun, etc.), and the seasons are 3-month periods (Dec - Feb, Mar - May, Jun - Aug, Sep - Nov). For the two waves that have to be split into seasons (e.g., December is lumped with January and February; June is lumped with July and August), we split the wave 50/50%, i.e, we make the assumption that half of the catch in that wave occurs in one season and the other half occurs in the next season.

```
# define seasons -----
```

```
table(d1$WAVE, useNA = "always")
```

```
0    1    2    3    4    5    6 <NA>
5  407  718 1230 1439 1095  581  944
```

```
d1$YRWAVE <- paste0(d1$YEAR, "_", d1$WAVE)
```

```
totw <- tapply(d1$LBSEST_SECWWT, list(d1$YRWAVE, d1$region, d1$sector), sum, na.rm = T)
```

```
f <- data.frame(cbind(totw[, , 1], totw[, , 2]))
```

```
names(f) <- c("Hire_FLK", "Hire_NCFL", "Hire_NNC", "Hire_VBM", "Rec_FLK", "Rec_NCFL", "Rec_
head(f)
```

	Hire_FLK	Hire_NCFL	Hire_NNC	Hire_VBM	Rec_FLK	Rec_NCFL	Rec_NNC
1981_2	NA	NA	NA	NA	2793285.5	270968.46	NA
1981_3	300873.764	NA	254049.420	NA	3915698.3	NA	NA
1981_4	50017.460	NA	2705.821	NA	905621.2	51537.26	NA
1981_5	3748.301	NA	NA	NA	479834.0	NA	NA
1981_6	NA	NA	NA	NA	848201.5	NA	NA
1981_NA	NA	74702.5	NA	1400.247	NA	NA	NA
	Rec_VBM						
1981_2	NA						
1981_3	NA						
1981_4	NA						
1981_5	NA						
1981_6	NA						
1981_NA	NA						

```
f$YEAR <- as.numeric(substr(rownames(f), 1, 4))
```

```
f$WAVE <- as.numeric(substr(rownames(f), 6, 8))
```

```
f # table with catches separated by wave and year
```

2 NOAA Fisheries recreational landings data

	Hire_FLK	Hire_NCFL	Hire_NNC	Hire_VBM	Rec_FLK
1981_2	NA	NA	NA	NA	2793285.48
1981_3	300873.7645	NA	254049.4202	NA	3915698.35
1981_4	50017.4599	NA	2705.8210	NA	905621.21
1981_5	3748.3009	NA	NA	NA	479833.98
1981_6	NA	NA	NA	NA	848201.53
1981_NA	NA	74702.5028	NA	1.400247e+03	NA
1982_1	NA	NA	NA	NA	478682.43
1982_2	84270.4684	NA	NA	NA	27122.56
1982_3	129622.4861	NA	NA	NA	3005422.81
1982_4	115226.2608	NA	27578.4417	NA	1993004.74
1982_5	13668.9931	NA	NA	NA	1699054.60
1982_6	4670.0135	NA	NA	NA	1219385.54
1982_NA	NA	92271.4232	NA	2.749224e+03	NA
1983_1	NA	NA	NA	NA	996045.13
1983_2	78251.8667	NA	NA	NA	817568.31
1983_3	296178.5757	215.4985	45363.4111	NA	5507616.65
1983_4	365077.5616	729.3286	NA	3.761981e+03	1066081.17
1983_5	258368.1069	NA	43327.1187	NA	1192916.67
1983_6	22129.7128	NA	NA	NA	420434.40
1983_NA	NA	40839.2141	NA	1.617195e+03	NA
1984_1	60534.3153	NA	NA	NA	496799.13
1984_2	109334.0510	NA	NA	NA	813248.44
1984_3	216520.5971	NA	NA	NA	688001.30
1984_4	532525.3239	21162.9274	NA	NA	568272.84
1984_5	61308.6459	4940.8922	NA	NA	231831.17
1984_6	40863.8371	NA	NA	NA	3918891.39
1984_NA	NA	48554.9922	NA	4.203048e+03	NA
1985_1	4462.0896	NA	NA	NA	468466.30
1985_2	127323.9916	NA	NA	NA	309953.53
1985_3	506974.7595	134535.9624	58514.1316	NA	2455502.04
1985_4	NA	7153.5135	NA	6.494967e+03	3224326.49
1985_5	4177.8282	NA	NA	NA	1419943.79
1985_6	74584.3285	NA	NA	NA	47109.12
1985_NA	NA	31300.1745	NA	2.385891e+03	NA
1986_1	3253.2455	NA	NA	NA	378536.05
1986_2	310022.8131	NA	7272.7549	NA	1383797.02
1986_3	984894.9967	5828.4550	637453.4573	3.033357e+02	1383618.58
1986_4	8737.1069	NA	114398.0335	5.948303e+04	845201.01
1986_5	46873.3986	9017.6689	54732.6542	NA	742559.70
1986_6	208572.4198	NA	3335.3577	NA	965598.77

2 NOAA Fisheries recreational landings data

1986_NA	NA	48709.7684	NA	1.871831e+03	NA
1987_1	137196.6816	NA	NA	NA	1074020.26
1987_2	18269.4109	NA	2210.4630	NA	2003339.20
1987_3	1180965.5743	8520.3829	164386.8683	NA	1915801.32
1987_4	388289.1983	1455.3872	581271.6432	5.137417e+03	1235019.50
1987_5	NA	2691.7381	101125.2712	1.728889e+04	134395.34
1987_6	49837.7886	NA	NA	NA	551009.18
1987_NA	NA	48576.0704	NA	2.541569e+03	NA
1988_1	26204.5928	NA	NA	NA	444795.17
1988_2	25034.6472	243.2488	NA	NA	1711171.70
1988_3	5870.0473	4631.1218	458976.9721	NA	1084220.85
1988_4	520698.5887	15358.7622	551295.8240	7.270123e+03	2476362.02
1988_5	143442.0053	8390.0712	104354.2524	8.492243e+04	617909.12
1988_6	105230.7224	259.1926	NA	NA	321463.76
1988_NA	NA	33510.4734	NA	2.483663e+03	NA
1989_1	199876.7079	NA	NA	NA	233799.65
1989_2	159188.4005	164384.2892	10972.6770	NA	4225325.37
1989_3	865296.2618	52054.9547	891309.5947	8.136501e+03	6530706.00
1989_4	1465737.2238	127173.4675	885492.2904	3.183983e+05	4519754.07
1989_5	83806.1572	12984.0005	180833.1940	1.183327e+05	1489025.07
1989_6	624798.7957	371.4887	2168.4765	NA	966437.14
1989_NA	NA	48171.8129	NA	6.610226e+03	NA
1990_1	54992.6978	NA	NA	NA	248606.45
1990_2	152128.6574	12579.4503	NA	NA	2405031.03
1990_3	30313.4970	125421.4842	552474.0996	NA	5538685.08
1990_4	558360.0137	36443.7626	1022506.7374	1.449129e+04	9345177.45
1990_5	36854.3280	9170.4562	174464.2228	7.455794e+03	1799169.61
1990_6	35923.3882	NA	520.4910	NA	564194.87
1990_NA	NA	94168.0119	NA	9.132357e+03	NA
1991_1	102043.3920	NA	NA	NA	540836.60
1991_2	379111.5134	937.2739	36535.6211	NA	1896921.08
1991_3	2302635.5002	141550.1050	604174.3059	NA	8237620.76
1991_4	1312391.3569	10843.8064	809450.3648	8.435581e+04	6319030.29
1991_5	93324.5228	1805.8554	256408.7253	5.776816e+04	5290624.40
1991_6	79267.4628	11638.9357	182.4796	NA	808010.50
1991_NA	NA	58776.6642	NA	1.704888e+04	NA
1992_1	471679.2124	NA	NA	NA	1774841.58
1992_2	436744.2277	7422.7732	18854.0712	NA	6185618.11
1992_3	1064259.1470	94786.8635	773610.1416	NA	6162233.87
1992_4	276194.5517	48764.1980	687836.6566	1.255492e+05	757141.83
1992_5	13809.2160	5532.8712	139384.1864	2.816470e+04	680152.92

2 NOAA Fisheries recreational landings data

1992_6	285721.0267	1140.4979	NA	NA	228864.68
1992_NA	NA	31867.1403	NA	7.435851e+03	NA
1993_1	90463.6572	NA	NA	NA	822313.03
1993_2	247911.9958	955.4524	30472.2897	NA	1215204.91
1993_3	842587.1612	206808.4794	731291.8750	NA	4492353.61
1993_4	98500.2702	146473.3962	661259.1516	7.136348e+05	1158634.88
1993_5	11059.1959	NA	132603.9764	1.427731e+04	1867851.33
1993_6	52571.2953	NA	222.8014	NA	410998.32
1993_NA	NA	37016.9977	NA	1.404324e+04	NA
1994_1	63525.8062	NA	NA	NA	614908.65
1994_2	57454.5525	40147.1522	20580.2350	NA	1848686.32
1994_3	166335.0996	325690.7931	1127590.0202	9.068972e+03	3860975.36
1994_4	510286.7047	34789.7114	425847.4537	5.523303e+05	1895392.91
1994_5	262700.1820	17298.8625	130286.8608	6.856423e+03	1133245.77
1994_6	71529.4042	154.8942	9677.4093	NA	179553.70
1994_NA	NA	19561.3056	NA	5.296000e+03	NA
1995_1	103240.8013	NA	NA	NA	740676.10
1995_2	853804.9029	54664.9994	10897.4618	NA	2637929.82
1995_3	1214448.2687	399179.4972	697448.0654	2.724721e+03	6843098.31
1995_4	431871.1571	167913.2635	814410.1056	1.064317e+05	3899582.86
1995_5	160263.5745	2703.5212	149576.2235	2.667280e+04	557184.96
1995_6	85062.9729	NA	1103.6279	8.113131e+01	880016.77
1995_NA	NA	43014.0167	NA	9.888226e+03	NA
1996_1	35208.0753	7846.4176	0.0000	NA	454438.34
1996_2	218752.8677	1768.0568	817.5057	NA	1034378.56
1996_3	2442321.7774	162044.9680	451630.5635	1.571826e+03	6221134.19
1996_4	131121.2140	74170.6704	401415.4483	4.659342e+04	1606830.44
1996_5	21696.9137	18720.2067	69033.4530	9.061178e+04	371044.76
1996_6	43335.4083	998.9381	231.8432	NA	886627.52
1996_NA	NA	44055.1611	NA	3.199404e+03	NA
1997_1	17079.3689	3183.6188	NA	NA	379586.51
1997_2	287832.9940	10057.2526	21667.7997	4.448883e+01	1328050.13
1997_3	2453169.8492	278685.3076	1033513.8821	2.761857e+03	12264482.97
1997_4	402896.5338	132744.9758	788661.1579	8.985193e+03	7529893.30
1997_5	58279.5995	37533.2441	282048.6226	9.832395e+04	439396.55
1997_6	52013.2925	4687.2662	3580.0003	NA	481846.64
1998_1	73087.7355	832.8979	0.0000	NA	271681.59
1998_2	297519.5348	7194.2556	4405.5371	1.160722e+02	3541176.70
1998_3	717661.5205	197051.4253	735203.5498	2.308373e+03	6312859.31
1998_4	312505.5033	38218.5532	918660.6667	7.541804e+05	1651975.25
1998_5	48482.1991	3465.9432	261878.8283	2.276001e+04	160611.24

2 NOAA Fisheries recreational landings data

1998_6	83203.5501	5857.6521	1685.4960	NA	640389.53
1999_1	97146.1032	13249.0415	NA	NA	1904555.74
1999_2	252008.9544	54096.3057	68229.8768	NA	1498377.50
1999_3	508649.1593	262921.9589	727564.3682	1.131979e+05	6764582.34
1999_4	195404.6702	100084.9281	1044438.7154	1.746593e+05	2993422.65
1999_5	68418.5697	21019.8335	173909.3357	6.067059e+02	308883.66
1999_6	116834.5768	6508.0368	609.9919	1.612885e+02	275312.67
2000_1	24738.6119	6532.3841	NA	NA	250739.40
2000_2	168806.2415	18491.9504	7444.5846	3.240762e+01	1456233.31
2000_3	524107.1827	163837.9514	1047963.8831	7.989470e+02	6175447.49
2000_4	464465.6055	139036.9399	1786682.4910	1.191410e+03	3270298.18
2000_5	211173.3520	32179.7861	297214.0115	9.740708e+02	1663716.32
2000_6	78190.3454	9474.3214	251.0068	1.975322e+01	563212.19
2001_1	206979.3316	9705.7248	NA	NA	453186.69
2001_2	292125.9142	46260.5820	11017.2008	7.142904e+00	1282139.55
2001_3	849789.7954	572194.0558	360412.6235	6.818656e+03	6293405.49
2001_4	808330.7146	423896.7585	1066496.7060	1.508136e+04	2442825.06
2001_5	66173.1946	35030.4250	199428.7925	1.240610e+04	793467.07
2001_6	138352.6869	3333.4528	1352.2216	2.197986e+01	425146.45
2002_1	55826.7807	1785.8554	0.0000	NA	345051.79
2002_2	299767.2785	71502.9893	27783.2046	5.138923e+01	1622268.74
2002_3	1043463.2842	1632381.5012	1954161.2216	1.321821e+05	2606943.71
2002_4	182534.9234	251742.3301	567978.2931	2.071477e+04	910613.27
2002_5	181662.3129	34198.7968	452882.2072	1.533410e+04	795613.17
2002_6	65664.1251	2167.5199	2367.9984	1.244497e+02	1824771.52
2003_1	84489.7923	2756.9405	NA	NA	543877.23
2003_2	130563.1573	40051.4990	37398.6274	1.699747e+01	1289231.54
2003_3	657331.8888	729571.0439	351393.3468	4.598796e+02	4719596.10
2003_4	275123.3687	167075.9356	498702.3583	1.420922e+04	2636140.58
2003_5	66406.2512	11228.9320	29076.1883	4.100432e+04	459396.21
2003_6	128424.6084	8312.7684	12323.4898	NA	646676.07
2004_1	102343.1102	321.9598	NA	NA	643883.91
2004_2	332642.4722	305327.0706	10906.8174	2.242078e+01	688723.05
2004_3	642567.4757	432048.6934	276168.2985	3.127000e+03	3120064.02
2004_4	332441.1474	13759.8362	755413.1322	1.137318e+05	1289299.85
2004_5	40166.3392	12518.5558	73884.8590	3.847026e+04	179995.30
2004_6	59499.9068	7844.5339	NA	1.593044e+02	359827.40
2005_1	56421.5819	2896.4255	NA	NA	343792.82
2005_2	94221.5573	118421.2282	1507.8607	NA	622983.27
2005_3	538518.5270	347902.2768	670947.2533	6.511947e+02	1716556.85
2005_4	113097.6370	255567.7726	2250098.8013	2.832497e+04	1216501.50

2 NOAA Fisheries recreational landings data

2005_5	29806.4290	5741.0552	229627.9171	9.044386e+03	75617.65
2005_6	26024.8665	4367.4433	1299.0535	NA	705757.77
2006_1	49456.9214	481.5067	NA	NA	418568.62
2006_2	131878.8301	15070.9844	5796.6298	3.492086e+01	2953455.31
2006_3	325792.3585	185401.2905	796673.6137	5.472015e+03	3508946.88
2006_4	131639.4927	106946.0818	1400395.7459	6.634345e+04	606432.81
2006_5	33510.7547	173514.3863	400946.6754	6.486219e+04	388387.08
2006_6	27811.1815	1922.8742	52427.5588	2.910072e+00	664641.11
2007_1	20430.0142	1253.2322	165.8849	NA	1355609.31
2007_2	148346.9410	59883.6053	14844.3015	1.388178e+02	1448285.20
2007_3	312828.9809	345248.6676	966460.8250	5.688771e+03	2193986.30
2007_4	172731.6760	94203.1609	1510649.0057	2.152043e+04	1951191.75
2007_5	32771.5859	1780.8230	25497.2931	5.753071e+04	926362.23
2007_6	31542.6603	4406.8079	813.1762	2.956369e+01	325052.74
2008_1	66366.3709	1231.6988	NA	NA	297413.25
2008_2	92615.8197	91079.4551	88553.4948	5.410718e+01	1327364.29
2008_3	375689.4952	1077085.1022	638562.0607	1.674707e+03	2183534.26
2008_4	126676.7826	24510.6284	783839.6556	9.586484e+04	2650100.59
2008_5	19438.7253	14550.6973	61498.5102	1.599922e+04	618778.59
2008_6	55426.0275	26842.1313	48412.1438	NA	1023358.65
2009_1	145422.1709	2114.4721	NA	NA	548871.47
2009_2	105560.5663	17395.3770	7506.7757	4.693716e+01	682265.88
2009_3	574115.8630	375002.6079	375407.6678	2.817377e+04	3417840.85
2009_4	116161.8296	56235.6591	446703.4979	6.744248e+04	598706.24
2009_5	31216.8500	2867.2594	93427.1551	1.059973e+04	386195.10
2009_6	44106.3576	4947.4404	5374.7627	0.000000e+00	1175567.25
2010_1	14106.4065	1183.6938	NA	NA	326412.60
2010_2	49161.1907	7758.5818	1451.3641	NA	209049.67
2010_3	340410.9199	433440.8039	976712.2918	2.277423e+03	1821535.72
2010_4	98575.4892	68943.3931	298696.1059	9.705147e+04	576787.52
2010_5	16596.9960	20491.3047	106585.8650	2.072151e+04	521366.15
2010_6	33229.8928	2853.2803	989.3009	NA	124439.00
2011_1	5836.9260	561.4132	NA	NA	96325.63
2011_2	96400.0980	7358.4198	3413.7822	2.551993e+02	607011.62
2011_3	375828.3691	149780.2324	817407.9573	1.989908e+03	2535780.70
2011_4	163696.1310	104922.2288	672637.2904	1.413401e+04	1573004.84
2011_5	39496.9624	9855.7916	81858.3829	1.465611e+03	769865.94
2011_6	87339.1714	3068.5688	1702.5462	NA	515061.11
2012_1	36821.2124	1426.8208	1091.5132	NA	325705.77
2012_2	137077.2847	37737.1615	44624.0825	NA	907680.62
2012_3	425790.2611	464859.0573	885255.2733	4.593218e+03	2479763.24

2 NOAA Fisheries recreational landings data

2012_4	129063.2472	44246.5803	210371.7456	1.306290e+04	673531.72
2012_5	66550.3469	64973.4374	93288.2799	4.231196e+03	929380.82
2012_6	60959.0401	2187.2815	1779.8681	NA	600081.94
2013_0	NA	NA	NA	6.437618e+03	NA
2013_1	40273.0429	685.7433	290.4661	NA	90480.96
2013_2	76014.4524	4791.2636	480.7299	3.771933e+01	537416.09
2013_3	150450.4869	305464.6059	168428.1882	1.153529e+03	5918763.88
2013_4	259379.9189	66703.4758	298703.6676	4.971875e+03	2683540.86
2013_5	31779.6669	14407.3832	171282.9326	3.464021e+03	350014.80
2013_6	56252.5362	2517.6379	1279.6204	NA	544582.41
2014_0	NA	NA	NA	3.608588e+04	NA
2014_1	10520.4506	3259.3964	NA	NA	353793.11
2014_2	57372.1480	5479.9568	2511.0227	2.031263e+01	2099748.55
2014_3	497966.7425	150213.3808	679220.5508	4.351780e+03	3975778.57
2014_4	229361.7142	34156.6244	146971.3189	1.597103e+04	1621326.41
2014_5	56743.2740	12970.2063	87233.8133	2.916618e+02	1460360.93
2014_6	45544.1094	4119.1926	759.2526	4.009366e+01	429012.86
2015_0	NA	NA	NA	1.748352e+05	NA
2015_1	89924.1637	2117.3775	NA	NA	36834.42
2015_2	272254.4974	9652.8461	60562.5789	1.424431e+01	1921333.40
2015_3	922470.7012	133668.6599	526282.5665	1.004099e+03	2330142.11
2015_4	210983.8295	12413.2463	624221.1835	2.305781e+04	5560719.14
2015_5	104619.5265	33776.8946	143688.6154	7.895665e+04	1677214.57
2015_6	129573.1497	4697.9438	20153.9625	NA	567684.23
2016_1	69649.9280	1349.5251	NA	NA	208103.89
2016_2	345342.4374	140442.7876	13431.0727	2.639211e+02	780971.95
2016_3	315996.5457	293561.0818	590465.4312	2.958322e+04	1875555.55
2016_4	83789.0091	8246.7891	359610.9766	1.511846e+05	1432462.61
2016_5	7982.4798	7539.9739	270901.1523	1.480608e+04	537933.12
2016_6	18902.1107	2913.3502	2157.1234	NA	167245.46
2017_1	2251.2393	583.4260	NA	NA	68363.67
2017_2	51034.2514	29595.1610	14556.5999	1.204006e+02	364071.77
2017_3	1030086.3486	94650.8142	569618.1211	3.769691e+03	1235624.61
2017_4	96754.8362	26694.7960	164125.2672	1.949092e+03	3073785.02
2017_5	24065.5921	8751.1280	80816.6336	2.533286e+04	491089.52
2017_6	23635.5797	912.9816	5964.2338	1.301596e+01	231958.35
2018_1	11472.9130	267.4297	NA	NA	137407.17
2018_2	44039.0548	1593.8892	6256.4127	1.149870e+02	1476572.22
2018_3	184156.6622	79757.1302	775249.4297	4.651212e+03	1995558.91
2018_4	73228.8542	14806.0747	362014.1087	3.109227e+04	2236679.62
2018_5	72317.4425	4987.5597	321095.4052	3.634558e+04	1573007.06

2 NOAA Fisheries recreational landings data

2018_6	16488.0967	3916.4019	472.2285	7.733737e+00	516503.80
2019_1	20082.6686	760.9299	NA	NA	238013.51
2019_2	157486.7280	12452.2622	25721.8869	NA	521964.88
2019_3	434125.1264	215818.6624	605973.8443	3.771762e+04	1182752.26
2019_4	209907.5499	16120.5455	214441.8613	4.427845e+04	2002350.22
2019_5	391814.5530	5005.6273	146797.0848	1.156134e+04	61952.84
2019_6	48410.4466	466.3692	5167.7689	1.612885e+02	57833.54
2020_1	NA	185.6236	NA	NA	29208.74
2020_2	9043.9102	4024.1289	6272.1824	NA	543554.92
2020_3	139285.5457	161242.6835	368302.7238	1.203957e+04	1123314.13
2020_4	439976.0290	25211.1659	201828.8119	9.142763e+03	774487.32
2020_5	297234.3672	33947.1059	90603.2405	3.042061e+04	245095.47
2020_6	24891.5786	883.0818	494.8315	1.821000e+01	191252.44
2021_1	15505.4306	321.9289	NA	NA	220137.23
2021_2	35095.4313	37409.0105	2043.7104	1.427543e+01	291276.82
2021_3	333171.4886	68394.4498	376553.7802	1.049071e+03	936836.16
2021_4	334601.5380	19773.6114	249480.9177	4.906534e+04	1353385.07
2021_5	499958.7353	15795.1271	242967.0387	5.086119e+04	718836.26
2021_6	26509.5669	5199.0236	715.4980	8.100330e+00	132124.10
2022_1	13136.0772	639.7193	NA	NA	97255.58
2022_2	30494.4008	6892.5164	8486.8910	NA	356620.00
2022_3	431004.1066	69373.8791	282923.0128	2.128348e+03	2228470.67
2022_4	85488.2771	8043.0385	46324.6662	1.066188e+04	460487.77
2022_5	50702.4924	5164.4181	87447.9009	5.723903e+03	646262.93
2022_6	16195.2651	391.1390	2770.8730	NA	39016.01
2023_1	6899.4172	113.6594	NA	NA	94317.83
2023_2	31601.1462	45397.1183	26776.7360	6.007665e+01	227368.50
2023_3	120806.3748	52677.7234	917576.6534	3.743502e+03	931385.61
2023_4	59490.9826	9801.4455	232969.4303	2.978505e+04	678063.87
2023_5	49697.8694	10045.8961	103931.7334	3.913939e+04	931220.84
2023_6	6036.6763	492.9613	NA	NA	183443.99
2024_1	791.1074	183.2757	NA	NA	82445.49
2024_2	82532.9838	25100.3010	5728.2128	8.438841e+01	215054.79
2024_3	281896.7265	36494.7437	265654.8738	4.029312e+03	1426097.26
2024_4	200628.0539	4812.0484	75322.4960	6.695225e+03	954109.75
2024_5	32206.1519	17813.6112	159633.5682	7.943312e+03	663746.22
2024_6	31272.2161	1168.2082	218.9170	NA	62773.96
	Rec_NCFL	Rec_NNC	Rec_VBM	YEAR	WAVE
1981_2	270968.461	NA	NA	1981	2
1981_3	NA	NA	NA	1981	3
1981_4	51537.259	NA	NA	1981	4

2 NOAA Fisheries recreational landings data

1981_5	NA	NA	NA	1981	5
1981_6	NA	NA	NA	1981	6
1981_NA	NA	NA	NA	1981	NA
1982_1	NA	NA	NA	1982	1
1982_2	NA	NA	NA	1982	2
1982_3	711098.009	42843.307	NA	1982	3
1982_4	386991.814	12094.937	18727.290	1982	4
1982_5	613744.843	23428.917	NA	1982	5
1982_6	NA	NA	NA	1982	6
1982_NA	NA	NA	NA	1982	NA
1983_1	NA	NA	NA	1983	1
1983_2	NA	NA	NA	1983	2
1983_3	20143.820	44226.032	NA	1983	3
1983_4	69834.733	NA	2565.809	1983	4
1983_5	29391.817	NA	2890.429	1983	5
1983_6	NA	NA	NA	1983	6
1983_NA	NA	NA	NA	1983	NA
1984_1	NA	NA	NA	1984	1
1984_2	NA	NA	NA	1984	2
1984_3	59564.622	NA	NA	1984	3
1984_4	426515.061	NA	NA	1984	4
1984_5	NA	NA	0.000	1984	5
1984_6	NA	NA	NA	1984	6
1984_NA	NA	NA	NA	1984	NA
1985_1	NA	NA	NA	1985	1
1985_2	NA	NA	NA	1985	2
1985_3	1118180.391	77477.096	NA	1985	3
1985_4	44527.998	NA	112944.274	1985	4
1985_5	NA	NA	NA	1985	5
1985_6	NA	NA	NA	1985	6
1985_NA	NA	NA	NA	1985	NA
1986_1	NA	NA	NA	1986	1
1986_2	NA	NA	NA	1986	2
1986_3	603584.417	0.000	NA	1986	3
1986_4	32067.516	2549.586	73390.216	1986	4
1986_5	74141.051	NA	30554.918	1986	5
1986_6	27079.033	NA	NA	1986	6
1986_NA	NA	NA	NA	1986	NA
1987_1	NA	NA	NA	1987	1
1987_2	NA	NA	NA	1987	2
1987_3	30116.500	NA	NA	1987	3

2 NOAA Fisheries recreational landings data

1987_4	92651.366	74122.360	42063.922	1987	4
1987_5	38859.186	5418.005	10075.850	1987	5
1987_6	NA	NA	NA	1987	6
1987_NA	NA	NA	NA	1987	NA
1988_1	NA	NA	NA	1988	1
1988_2	NA	NA	NA	1988	2
1988_3	54755.955	23711.146	NA	1988	3
1988_4	222834.481	179057.804	29325.047	1988	4
1988_5	43432.685	NA	4241.116	1988	5
1988_6	NA	NA	NA	1988	6
1988_NA	NA	NA	NA	1988	NA
1989_1	139949.965	NA	NA	1989	1
1989_2	111701.985	10512.208	NA	1989	2
1989_3	227363.510	272095.085	12546.840	1989	3
1989_4	130611.175	54238.048	675840.429	1989	4
1989_5	53101.278	651.014	308558.121	1989	5
1989_6	417223.578	1875.212	NA	1989	6
1989_NA	NA	NA	NA	1989	NA
1990_1	NA	NA	NA	1990	1
1990_2	NA	1916.750	NA	1990	2
1990_3	98533.276	23762.838	NA	1990	3
1990_4	100213.515	89871.470	146908.507	1990	4
1990_5	113586.480	2988.903	362516.474	1990	5
1990_6	NA	NA	NA	1990	6
1990_NA	NA	NA	NA	1990	NA
1991_1	NA	NA	NA	1991	1
1991_2	NA	132457.227	NA	1991	2
1991_3	372647.616	111487.795	NA	1991	3
1991_4	73235.386	18973.369	313639.776	1991	4
1991_5	11784.127	14471.255	151146.958	1991	5
1991_6	NA	NA	NA	1991	6
1991_NA	NA	NA	NA	1991	NA
1992_1	NA	NA	NA	1992	1
1992_2	NA	NA	NA	1992	2
1992_3	92148.045	41376.379	20526.808	1992	3
1992_4	126696.827	24302.047	76676.416	1992	4
1992_5	26550.880	23482.434	412094.037	1992	5
1992_6	NA	NA	NA	1992	6
1992_NA	NA	NA	NA	1992	NA
1993_1	NA	NA	NA	1993	1
1993_2	89986.989	NA	NA	1993	2

2 NOAA Fisheries recreational landings data

1993_3	370532.453	33545.453	6991.506	1993	3
1993_4	250305.146	259128.423	695338.082	1993	4
1993_5	13889.315	3725.324	169619.013	1993	5
1993_6	18027.925	NA	NA	1993	6
1993_NA	NA	NA	NA	1993	NA
1994_1	28223.840	NA	NA	1994	1
1994_2	46344.061	NA	NA	1994	2
1994_3	963122.439	220282.038	4547.236	1994	3
1994_4	151284.747	20987.542	509707.555	1994	4
1994_5	136265.383	21835.365	80212.622	1994	5
1994_6	383701.640	1802.248	NA	1994	6
1994_NA	NA	NA	NA	1994	NA
1995_1	NA	NA	NA	1995	1
1995_2	480461.867	NA	NA	1995	2
1995_3	804233.833	225758.606	160479.740	1995	3
1995_4	170952.186	26748.450	448822.879	1995	4
1995_5	37780.398	18935.018	NA	1995	5
1995_6	56707.695	NA	NA	1995	6
1995_NA	NA	NA	NA	1995	NA
1996_1	NA	NA	NA	1996	1
1996_2	314628.146	20209.360	NA	1996	2
1996_3	625690.901	87985.514	NA	1996	3
1996_4	111818.582	63663.755	408494.551	1996	4
1996_5	69800.214	24720.906	77611.636	1996	5
1996_6	NA	NA	NA	1996	6
1996_NA	NA	NA	NA	1996	NA
1997_1	NA	NA	NA	1997	1
1997_2	347295.859	63503.937	NA	1997	2
1997_3	687821.842	219639.986	NA	1997	3
1997_4	117886.588	326036.328	149860.531	1997	4
1997_5	163498.444	51077.591	48011.431	1997	5
1997_6	NA	NA	NA	1997	6
1998_1	NA	NA	NA	1998	1
1998_2	57196.548	NA	NA	1998	2
1998_3	260289.770	164723.993	NA	1998	3
1998_4	38810.562	199018.669	470816.565	1998	4
1998_5	28150.625	48561.629	135952.733	1998	5
1998_6	235376.551	NA	NA	1998	6
1999_1	28295.456	NA	NA	1999	1
1999_2	161859.944	7907.629	NA	1999	2
1999_3	998925.562	205305.614	51453.808	1999	3

2 NOAA Fisheries recreational landings data

1999_4	220964.231	663226.036	215387.059	1999	4
1999_5	91326.076	438366.095	256782.821	1999	5
1999_6	46846.176	NA	NA	1999	6
2000_1	NA	NA	NA	2000	1
2000_2	292817.152	NA	NA	2000	2
2000_3	1860759.936	617314.126	NA	2000	3
2000_4	362528.918	173566.777	1440372.603	2000	4
2000_5	220496.767	36099.433	168106.317	2000	5
2000_6	47821.441	NA	NA	2000	6
2001_1	280396.645	NA	NA	2001	1
2001_2	286154.324	NA	NA	2001	2
2001_3	2681450.090	881436.853	119550.061	2001	3
2001_4	186070.834	865658.069	228166.886	2001	4
2001_5	99831.517	13626.222	24906.326	2001	5
2001_6	80466.902	NA	NA	2001	6
2002_1	0.000	0.000	NA	2002	1
2002_2	72986.906	33095.271	NA	2002	2
2002_3	1394867.652	363688.365	319894.863	2002	3
2002_4	261541.132	389650.618	446128.499	2002	4
2002_5	434852.883	85496.406	966332.674	2002	5
2002_6	319697.989	NA	NA	2002	6
2003_1	39787.277	NA	NA	2003	1
2003_2	110441.284	62461.897	NA	2003	2
2003_3	818832.674	353696.442	NA	2003	3
2003_4	786858.585	432922.024	122391.084	2003	4
2003_5	8290.240	19119.891	782389.474	2003	5
2003_6	96225.360	NA	NA	2003	6
2004_1	7766.988	NA	NA	2004	1
2004_2	200018.750	435713.032	NA	2004	2
2004_3	780680.858	80535.729	NA	2004	3
2004_4	309322.339	72736.412	128301.965	2004	4
2004_5	5193.736	NA	58236.601	2004	5
2004_6	51321.055	4272.544	NA	2004	6
2005_1	NA	NA	NA	2005	1
2005_2	555997.501	99873.420	NA	2005	2
2005_3	1162230.575	319288.797	NA	2005	3
2005_4	149887.815	431042.453	275682.187	2005	4
2005_5	20126.383	54155.540	213554.264	2005	5
2005_6	10703.630	NA	NA	2005	6
2006_1	NA	NA	NA	2006	1
2006_2	185852.110	NA	NA	2006	2

2 NOAA Fisheries recreational landings data

2006_3	1198919.353	213816.974	222334.322	2006	3
2006_4	249267.647	501064.507	571986.623	2006	4
2006_5	27558.269	22774.847	420176.095	2006	5
2006_6	90256.116	11885.085	NA	2006	6
2007_1	113126.281	NA	NA	2007	1
2007_2	79674.506	NA	NA	2007	2
2007_3	1784221.243	982353.036	NA	2007	3
2007_4	376988.429	253437.399	208924.036	2007	4
2007_5	29406.607	79811.644	27845.086	2007	5
2007_6	NA	168028.510	NA	2007	6
2008_1	26690.282	NA	NA	2008	1
2008_2	666548.746	NA	NA	2008	2
2008_3	808463.272	143008.342	44964.096	2008	3
2008_4	102477.513	80556.843	44115.250	2008	4
2008_5	14935.294	NA	7739.115	2008	5
2008_6	29547.066	NA	NA	2008	6
2009_1	9241.668	NA	NA	2009	1
2009_2	1030753.438	465640.888	NA	2009	2
2009_3	2030652.102	1244129.669	7087.781	2009	3
2009_4	673984.863	642562.676	1570414.278	2009	4
2009_5	11403.457	6844.584	30416.945	2009	5
2009_6	49092.147	NA	NA	2009	6
2010_1	NA	22325.895	NA	2010	1
2010_2	1518161.094	NA	NA	2010	2
2010_3	1457986.856	808715.818	11260.876	2010	3
2010_4	150645.041	249105.267	505803.531	2010	4
2010_5	12406.577	30332.751	200346.451	2010	5
2010_6	NA	NA	NA	2010	6
2011_1	34555.638	NA	NA	2011	1
2011_2	107923.039	NA	NA	2011	2
2011_3	866645.103	939778.302	NA	2011	3
2011_4	484504.029	600021.700	2626410.052	2011	4
2011_5	157468.639	523515.640	NA	2011	5
2011_6	NA	NA	NA	2011	6
2012_1	NA	2302.064	NA	2012	1
2012_2	1396780.799	20740.405	NA	2012	2
2012_3	995274.115	738963.571	NA	2012	3
2012_4	376987.007	121312.497	1041694.282	2012	4
2012_5	98557.164	153042.474	NA	2012	5
2012_6	NA	46246.051	NA	2012	6
2013_0	NA	NA	NA	2013	0

2 NOAA Fisheries recreational landings data

2013_1	21440.509	NA	NA	2013	1
2013_2	35854.832	NA	NA	2013	2
2013_3	1676248.628	246781.020	NA	2013	3
2013_4	295827.837	286021.569	137491.131	2013	4
2013_5	34884.794	7874.372	227001.156	2013	5
2013_6	NA	NA	NA	2013	6
2014_0	NA	NA	NA	2014	0
2014_1	NA	NA	NA	2014	1
2014_2	33933.384	12677.929	NA	2014	2
2014_3	987984.218	292367.815	290639.196	2014	3
2014_4	141508.489	964223.136	1600731.334	2014	4
2014_5	103701.776	20435.535	148959.019	2014	5
2014_6	5349.406	NA	NA	2014	6
2015_0	NA	NA	NA	2015	0
2015_1	NA	NA	NA	2015	1
2015_2	119360.670	48147.133	NA	2015	2
2015_3	844699.884	2026488.066	536501.660	2015	3
2015_4	383767.988	1050322.063	2011174.244	2015	4
2015_5	200599.935	380237.481	2098120.345	2015	5
2015_6	NA	NA	NA	2015	6
2016_1	35519.395	NA	NA	2016	1
2016_2	665307.128	23051.131	NA	2016	2
2016_3	3492533.874	1154898.312	546137.274	2016	3
2016_4	134696.856	470319.918	1394689.356	2016	4
2016_5	28918.336	56392.308	245300.042	2016	5
2016_6	15647.057	NA	NA	2016	6
2017_1	13461.888	NA	NA	2017	1
2017_2	162822.865	6205.602	NA	2017	2
2017_3	1965899.683	637939.840	50758.107	2017	3
2017_4	138289.429	148993.679	1262079.713	2017	4
2017_5	12138.232	8185.031	513985.029	2017	5
2017_6	5317.948	NA	NA	2017	6
2018_1	45330.868	NA	NA	2018	1
2018_2	181104.623	NA	NA	2018	2
2018_3	2066383.613	712987.295	11894.035	2018	3
2018_4	259620.647	322678.281	888642.213	2018	4
2018_5	52628.105	312058.626	1970588.163	2018	5
2018_6	1023.673	NA	NA	2018	6
2019_1	17540.107	NA	NA	2019	1
2019_2	478600.959	NA	NA	2019	2
2019_3	1447710.287	784303.051	211810.531	2019	3

2 NOAA Fisheries recreational landings data

2019_4	135802.575	694820.774	757784.150	2019	4
2019_5	42946.029	208116.039	470826.260	2019	5
2019_6	9883.151	NA	NA	2019	6
2020_1	NA	NA	NA	2020	1
2020_2	306892.361	NA	NA	2020	2
2020_3	1572309.784	871138.027	14858.043	2020	3
2020_4	84716.506	208713.607	462496.274	2020	4
2020_5	61174.394	9393.537	211745.154	2020	5
2020_6	27128.921	NA	NA	2020	6
2021_1	NA	NA	NA	2021	1
2021_2	197020.341	NA	NA	2021	2
2021_3	820299.736	522426.815	25610.586	2021	3
2021_4	71258.533	74245.665	637419.419	2021	4
2021_5	252799.591	NA	418259.071	2021	5
2021_6	87130.105	NA	1979.736	2021	6
2022_1	NA	NA	NA	2022	1
2022_2	129995.246	NA	NA	2022	2
2022_3	2274049.909	70124.055	50620.640	2022	3
2022_4	261629.144	74508.172	596803.936	2022	4
2022_5	35945.538	66762.310	558853.748	2022	5
2022_6	NA	NA	NA	2022	6
2023_1	27164.606	NA	NA	2023	1
2023_2	187646.403	NA	NA	2023	2
2023_3	2223128.761	50810.398	6212.940	2023	3
2023_4	96249.184	124378.825	431039.117	2023	4
2023_5	105256.134	NA	435238.829	2023	5
2023_6	NA	NA	NA	2023	6
2024_1	NA	NA	NA	2024	1
2024_2	25728.471	NA	NA	2024	2
2024_3	1145791.778	45630.340	1476.410	2024	3
2024_4	120926.013	60676.938	945017.750	2024	4
2024_5	NA	NA	400112.682	2024	5
2024_6	66072.800	NA	NA	2024	6

```
f[is.na(f)] <- 0
```

```
f <- f[which(f$YEAR >= 1985), ]
```

```
f <- f[which(f$YEAR <= 2024), ]
```

```
table(f$YEAR)
```

2 NOAA Fisheries recreational landings data

1985	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
7	7	7	7	7	7	7	7	7	7	7	7	6	6	6	6
2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016
6	6	6	6	6	6	6	6	6	6	6	6	7	7	7	6
2017	2018	2019	2020	2021	2022	2023	2024								
6	6	6	6	6	6	6	6								

```
table(f$WAVE)
```

```
0  1  2  3  4  5  6
15 40 40 40 40 40 40
```

```
# deal with NAs in many locations -----
# calculate percentages of catch by wave, for each region and fleet

avw <- tapply(d1$LBSEST_SECWWT, list(d1$WAVE, d1$region, d1$sector), sum, na.rm = T)
avw <- avw[-1, , ]
avw <- data.frame(cbind(avw[, , 1], avw[, , 2]))
names(avw) <- c("Hire_FLK", "Hire_NCFL", "Hire_NNC", "Hire_VBM", "Rec_FLK", "Rec_NCFL", "Rec_NNC")
avwp <- apply(avw, 2, function (x) x / sum(x, na.rm = T))
#colSums(avwp)

# go through each column, find NAs and fill in waves by percentages

pre_colsum <- colSums(f, na.rm = T) # column sums to check later

lis <- which(f$WAVE == 0) # list of rows with NA wave data

for (i in 1:8) { # loop through each column
  for(j in lis) { # loop through list of NA waves
    if (f[j, i] > 0) { # if there are NA wave catches, split by known proportion from other waves
      miscatch <- avwp[, i] * f[j, i] # missing catch is known proportion * NA wave catch
      f[which(f$YEAR == f$YEAR[j] & f$WAVE != 0), i] <- f[which(f$YEAR == f$YEAR[j] & f$WAVE != 0), i] + miscatch
      f[j, i] <- 0 # set to zero after catch is attributed to waves
    }
  }
}
```


2 NOAA Fisheries recreational landings data

```
f[lis, 1:8] # should be all zeros now
```

	Hire_FLK	Hire_NCFL	Hire_NNC	Hire_VBM	Rec_FLK	Rec_NCFL	Rec_NNC	Rec_VBM
1985_NA	0	0	0	0	0	0	0	0
1986_NA	0	0	0	0	0	0	0	0
1987_NA	0	0	0	0	0	0	0	0
1988_NA	0	0	0	0	0	0	0	0
1989_NA	0	0	0	0	0	0	0	0
1990_NA	0	0	0	0	0	0	0	0
1991_NA	0	0	0	0	0	0	0	0
1992_NA	0	0	0	0	0	0	0	0
1993_NA	0	0	0	0	0	0	0	0
1994_NA	0	0	0	0	0	0	0	0
1995_NA	0	0	0	0	0	0	0	0
1996_NA	0	0	0	0	0	0	0	0
2013_0	0	0	0	0	0	0	0	0
2014_0	0	0	0	0	0	0	0	0
2015_0	0	0	0	0	0	0	0	0

```
pos_colsum <- colSums(f, na.rm = T)
```

```
pre_colsum == pos_colsum # check that catch was parsed correctly - column sums should be i
```

Hire_FLK	Hire_NCFL	Hire_NNC	Hire_VBM	Rec_FLK	Rec_NCFL	Rec_NNC	Rec_VBM
TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
YEAR	WAVE						
TRUE	TRUE						

```
f <- f[which(f$WAVE != 0), ] # remove NA waves which now contain no catch
```

```
# split waves 3 and 6 50/50% to parse into seasons -----
```

```
lis <- which(f$WAVE == 3 | f$WAVE == 6) # list of 3 and 6 waves for all years
```

```
for (i in lis) {
  # go through list of waves
  temp <- f[i, ] / 2 # split catch in half
  temp[9] <- f$YEAR[i] # reformat year and wave data columns
  temp[10] <- f$WAVE[i]
  temp <- rbind(temp, temp) # create two half-waves and relabel with new wave reference n
```

2 NOAA Fisheries recreational landings data

```
temp[1, 10] <- temp[1, 10] - 0.5
temp[2, 10] <- temp[2, 10] + 0.5
f <- rbind(f, temp)      # concatenate with main data table
}

f <- f[-lis, ] # remove waves that have been split in half

pos_colsum <- colSums(f, na.rm = T)
pre_colsum == pos_colsum # check that catch was parsed correctly - column sums should be i
```

Hire_FLK	Hire_NCFL	Hire_NNC	Hire_VBM	Rec_FLK	Rec_NCFL	Rec_NNC	Rec_VBM
TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE
YEAR	WAVE						
FALSE	FALSE						

```
f <- f[order(f$YEAR, f$WAVE), ] # reorder table in chronological order

f$YEAR[which(f$WAVE == 6.5)] <- f$YEAR[which(f$WAVE == 6.5)] + 1 # add year to December wa
f$WAVE[which(f$WAVE == 6.5)] <- 0.5 # change December wave to small number so it gets summ

table(f$WAVE)
```

0.5	1	2	2.5	3.5	4	5	5.5
40	40	40	40	40	40	40	40

```
f$seas <- cut(f$WAVE, breaks = c(0, 1.5, 3, 4.5, 6))
table(f$seas)
```

(0,1.5]	(1.5,3]	(3,4.5]	(4.5,6]
80	80	80	80

```
f$seas1 <- NA
f$seas1[which(f$seas == "(0,1.5)")] <- "DJF" # DJF is 0.5 and 1
f$seas1[which(f$seas == "(1.5,3)")] <- "MAM" # MAM is 2 and 2.5
f$seas1[which(f$seas == "(3,4.5)")] <- "JJA" # JJA is 3.5 and 4
f$seas1[which(f$seas == "(4.5,6)")] <- "SON" # SON is 5 and 5.5
```

2 NOAA Fisheries recreational landings data

```
temp <- tapply(f[, 1], list(f$seas1, f$YEAR), sum, na.rm = T)
temp1 <- matrix(temp)

for (i in 2:8) {
  temp2 <- matrix(tapply(f[, i], list(f$seas1, f$YEAR), sum, na.rm = T))
  temp1 <- cbind(temp1, temp2)
}

findat <- data.frame(sort(rep(as.numeric(colnames(temp)), 4)),
                     rep(rownames(temp), ncol(temp)), temp1)
names(findat) <- c("year", "season", names(f)[1:8])

colSums(f[, 1:8], na.rm = T) == colSums(findat[, 3:10], na.rm = T) # check that total catch
```

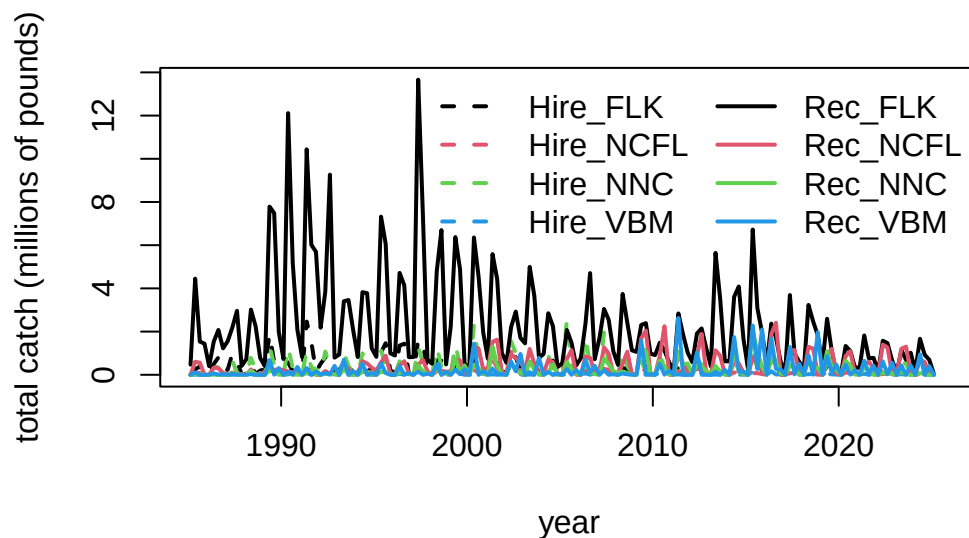
Hire_FLK	Hire_NCFL	Hire_NNC	Hire_VBM	Rec_FLK	Rec_NCFL	Rec_NNC	Rec_VBM
TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE	TRUE

```
findat$yrseas <- findat$year + as.numeric(as.factor(findat$season))/4-0.125
```

We plot the results by region and sector, over the time series of year and season. We can clearly see the seasonality in the catch by region.

```
matplot(findat$yrseas, findat[, 3:10]/10^6, lty = c(rep(2, 4), rep(1, 4)), col = rep(1:4, 2),
        xlab = "year", ylab = "total catch (millions of pounds)")
legend("topright", colnames(findat[, 3:10]), lty = c(rep(2, 4), rep(1, 4)), col = rep(1:4,
```

2 NOAA Fisheries recreational landings data



```
# check that totals match
colSums(findat[, 3:10], na.rm = T)
```

Hire_FLK	Hire_NCFL	Hire_NNC	Hire_VBM	Rec_FLK	Rec_NCFL	Rec_NNC	Rec_VBM
55604999	16909916	57944903	5836953	371462933	70803866	32556596	40081529

```
colSums(fin[-c(1:5), ], na.rm = T)
```

Hire_FLK	Hire_NCFL	Hire_NNC	Hire_VBM	Rec_FLK	Rec_NCFL	Rec_NNC	Rec_VBM
54887476	16736926	57886388	5828072	363537632	69641157	32479119	39968585

```
# the values should be close but slightly >1 because the last data set includes December 19
colSums(findat[, 3:10], na.rm = T) / colSums(fin[-c(1:5), ], na.rm = T)
```

Hire_FLK	Hire_NCFL	Hire_NNC	Hire_VBM	Rec_FLK	Rec_NCFL	Rec_NNC	Rec_VBM
1.013073	1.010336	1.001011	1.001524	1.021800	1.016696	1.002385	1.002826

2 NOAA Fisheries recreational landings data

```
# very tiny differences
```

3 Analysis of international landings estimates from Sea Around Us database

Here we will explore commercial and recreational landings data as estimated by the Sea Around Us Project. This data set includes reported data for fisheries worldwide as well as reconstructed estimates for unreported landings. For each country, project scientists assemble available data on landings and use additional sociological and fishing data plus expert information and opinion, to produce their best estimates of total landings from all fishing sectors and for all species globally.

3.1 Data access and upload

For catches within each country's exclusive economic zones, data are downloaded from the Sea Around Us site (Pauly, Zeller, and Palomares, 2020.). In the Biodiversity by Taxon Tools and Data, we filter by EEZ for All regions, and then filter by Taxon at the Species level for Common dolphinfish (*Coryphaena hippurus*). After clicking "View graphs" one can access the full data set using the "Download Data" button. The high seas data are accessed in the Basic Search High seas section of the site. The individual regions can be clicked on (e.g., Atlantic, Western Central) and then the data are downloaded using the Download Data button.

The current version of data in use is version 50.1, accessed June 2024 (the data files can be found on github in the SEFSC-dolphin-analyses/data /SAU/ folder). This was still the most recent version of data available as of June 2025.

3.2 Read in the high seas landings

First we read in the high seas landings for the Western Central Atlantic and Northwest Atlantic zones. These are equivalent to FAO areas 21 and 31.

3 Analysis of international landings estimates from Sea Around Us database

```
# clear workspace
rm(list = ls())

if(!require("pals")) install.packages("pals")
library(pals)

# find data files
hslis <- dir("data/SAU/")[grep("HighSeas", dir("data/SAU/"))]

# identify areas 21 and 31
hslis[1]; hslis[3]
```

```
[1] "SAU HighSeas 21 v50-1.csv"
```

```
[1] "SAU HighSeas 31 v50-1.csv"
```

```
#download data
h1 <- read.csv(paste0("data/SAU/", hslis[1]))
h3 <- read.csv(paste0("data/SAU/", hslis[3]))

# confirm correct areas
#unique(h1$area_name)
#unique(h3$area_name)

hs <- rbind(h1, h3)
```

We extract the data for dolphinfish and look at the characteristics of the landings. Most of the landings are in the Western Central Atlantic, and the primary fishing entities are Venezuela and Saint Lucia. The fishing sector is all industrial, and the vast majority of the catch is reported landings, with only a tiny fraction of discards or unreported landings listed in the database. The gear type is primarily long line.

```
# check species name and filter out dolphinfish
unique(hs$common_name)[grep("dolphin", unique(hs$common_name))]
```

```
[1] "Common dolphinfish"
```

3 Analysis of international landings estimates from Sea Around Us database

```
hs <- hs[which(hs$common_name == "Common dolphinfish"), ]  
  
# look at the categories within each column of data  
apply(hs[1:13], 2, table)
```

\$area_name

Atlantic, Northwest Atlantic, Western Central
113 160

\$area_type

high_seas
273

\$year

1985	1986	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012
1	1	1	1	1	1	1	6	4	1	5	10	5	8	10	12
2013	2014	2015	2016	2017	2018	2019									
24	22	30	33	33	29	34									

\$scientific_name

Coryphaena hippurus
273

\$common_name

Common dolphinfish
273

\$functional_group

Large pelagics (≥ 90 cm)
273

\$commercial_group

Perch-likes

3 Analysis of international landings estimates from Sea Around Us database

273

\$fishing_entity

Barbados	Belize
6	3
Brazil	Canada
19	18
Guyana	Panama
1	1
Saint Lucia	Saint Vincent & the Grenadines
5	28
Spain	Suriname
80	1
Trinidad & Tobago	USA
14	74
Venezuela	
23	

\$fishing_sector

Industrial

273

\$catch_type

Discards Landings

5 268

\$reporting_status

Reported Unreported

253 20

\$gear_type

artisanal fishing gear	bottom trawl	gillnet
7	9	11
hand lines	longline	mixed gear
17	187	1
pole and line	pots or traps	unknown by source

3 Analysis of international landings estimates from Sea Around Us database

19

9

13

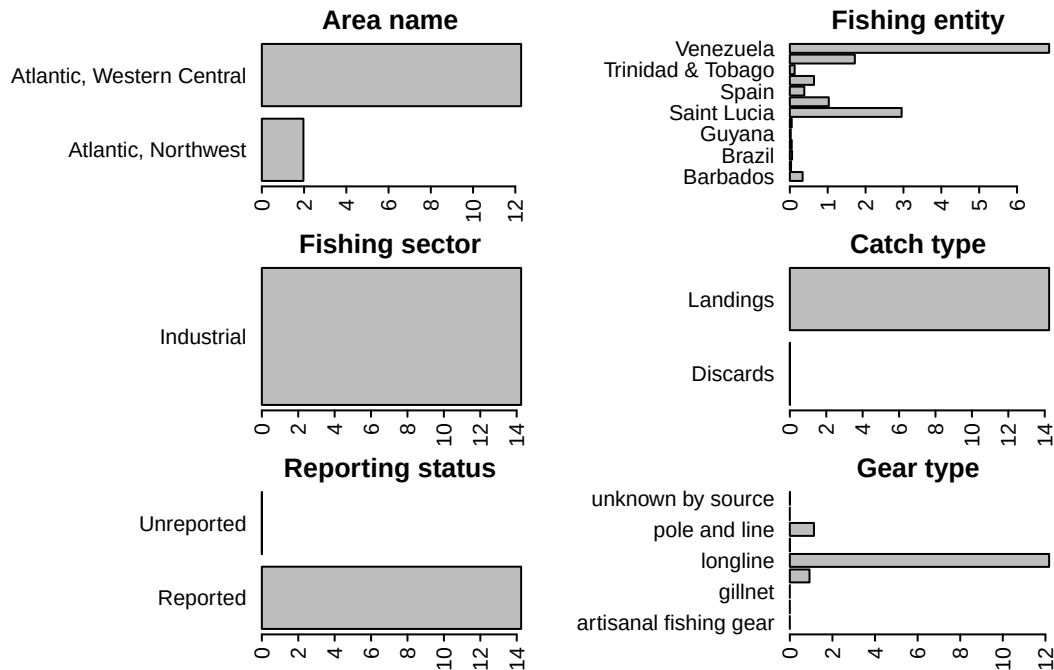
\$end_use_type

	Direct human consumption	Fishmeal and fish oil
5	194	37
Other		
37		

```
# convert from metric tonnes to pounds
hs$pounds <- hs$tonnes * 1000 * 2.20462

# produce barplot of key data fields
par(mar = c(3, 20, 3, 1), mfrow = c(3, 2), mex = 0.5)
barplot(tapply(hs$pounds/10^6, hs$area_name, sum, na.rm = T), las = 2, xlab = "millions of
#barplot(tapply(hs$pounds/10^6, hs$area_type, sum, na.rm = T), las = 2, ylab = "millions of
#barplot(tapply(hs$pounds/10^6, hs$year, sum, na.rm = T), las = 2, ylab = "millions of poun
barplot(tapply(hs$pounds/10^6, hs$fishing_entity, sum, na.rm = T), las = 2, xlab = "million
barplot(tapply(hs$pounds/10^6, hs$fishing_sector, sum, na.rm = T), las = 2, xlab = "million
barplot(tapply(hs$pounds/10^6, hs$catch_type, sum, na.rm = T), las = 2, xlab = "millions of
barplot(tapply(hs$pounds/10^6, hs$reporting_status, sum, na.rm = T), las = 2, xlab = "milli
barplot(tapply(hs$pounds/10^6, hs$gear_type, sum, na.rm = T), las = 2, xlab = "millions of
```

3 Analysis of international landings estimates from Sea Around Us database



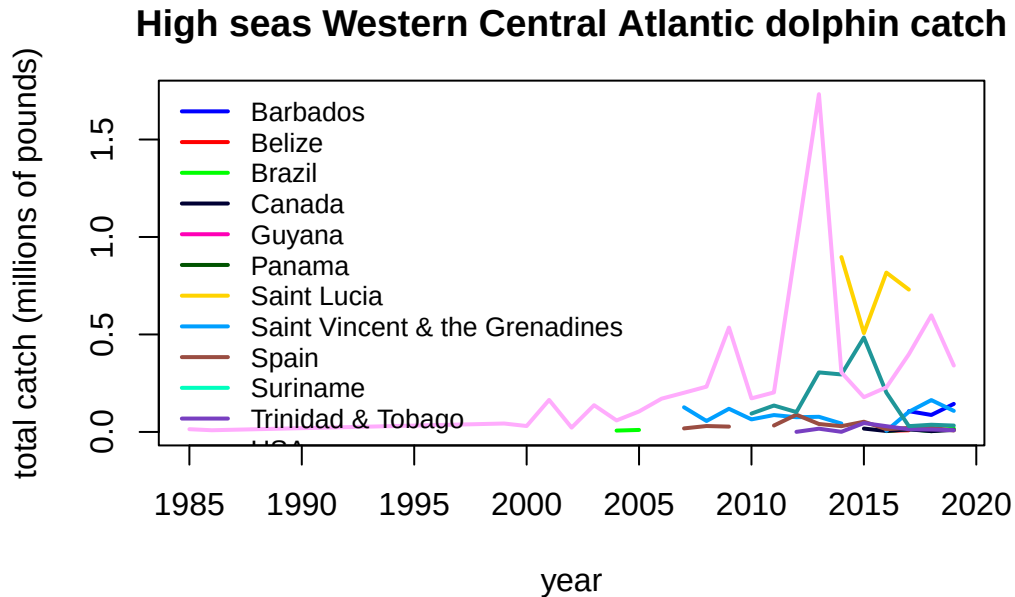
3.3 Summarize the high seas landings

We plot the annual catches by fishing entity to look at trends over time.

```
hs$fishing_entity <- as.factor(hs$fishing_entity)
hs$fishing_entity <- droplevels(hs$fishing_entity)

par(mar = c(5, 5, 3, 1), mfrow = c(1, 1))
tab <- tapply(hs$pounds, list(hs$year, hs$fishing_entity), sum, na.rm = T)
matplot(as.numeric(rownames(tab)), tab/10^6, type = "l", col = glasbey(ncol(tab)), lty = 1,
        main = "High seas Western Central Atlantic dolphin catch")
legend("topleft", colnames(tab), col = glasbey(ncol(tab)), lty = 1, lwd = 2, cex = 0.8, bty
```

3 Analysis of international landings estimates from Sea Around Us database



Finally we output the subset of high seas dolphin catch for the two areas. Because we have already accounted for the USA landings in our national reporting and are aiming to only summarize international landings here, we filter out the USA landings from the Sea Around Us database.

```
hs <- hs[which(hs$fishing_entity != "USA"), ]

hs$year <- factor(hs$year, levels = 1950:2019)
tab <- tapply(hs$pounds, list(hs$year, hs$area_name), sum, na.rm = T)
tab[is.na(tab)] <- 0
tail(tab, 40)
```

	Atlantic, Northwest Atlantic	Atlantic, Western Central
1980	0.000	0.000
1981	0.000	0.000
1982	0.000	0.000
1983	0.000	0.000
1984	0.000	0.000
1985	0.000	14037.129
1986	0.000	8748.399
1987	0.000	0.000

3 Analysis of international landings estimates from Sea Around Us database

1988	0.000	0.000
1989	0.000	0.000
1990	0.000	0.000
1991	0.000	0.000
1992	0.000	0.000
1993	0.000	0.000
1994	0.000	0.000
1995	0.000	0.000
1996	0.000	0.000
1997	0.000	0.000
1998	0.000	0.000
1999	0.000	43168.296
2000	0.000	30284.627
2001	0.000	164272.223
2002	0.000	22147.477
2003	0.000	137103.181
2004	0.000	66058.258
2005	0.000	115171.863
2006	0.000	170522.101
2007	14687.527	330838.620
2008	28552.636	340863.719
2009	18719.121	662720.661
2010	4510.085	232093.837
2011	21589.424	300498.224
2012	72398.812	1059210.392
2013	34741.604	2471791.396
2014	27922.953	1247462.983
2015	90313.638	710059.125
2016	15319.433	1094893.891
2017	34806.933	1374259.739
2018	29328.358	885814.754
2019	21870.621	632747.157

```
# write summary to merge with EEZ data
write.csv(tab, file = "data/SAU/high_seas_by_year.csv")
```

3.4 Read in the landings within exclusive economic zones (EEZs)

Now we read in the landings reported by each country or jurisdiction's respective EEZ. Because we have extracted dolphin landings for EEZs worldwide, we need to separate out the landings for the Atlantic Ocean basin.

```
rm(list = ls())
d <- read.csv("data/SAU/SAU Taxa 600006 v50-1.csv", header = T, sep = ",") # v. 50.1 - Jun
head(d)
```

	area_name	area_type	year	scientific_name	common_name
1	Australia	eez	1950	Coryphaena hippurus	Common dolphinfish
2	Australia	eez	1950	Coryphaena hippurus	Common dolphinfish
3	Australia	eez	1950	Coryphaena hippurus	Common dolphinfish
4	Bahamas	eez	1950	Coryphaena hippurus	Common dolphinfish
5	Barbados	eez	1950	Coryphaena hippurus	Common dolphinfish
6	Bermuda (UK)	eez	1950	Coryphaena hippurus	Common dolphinfish
	functional_group	commercial_group	fishing_entity	fishing_sector	
1	Large pelagics (>=90 cm)	Perch-likes	Australia	Industrial	
2	Large pelagics (>=90 cm)	Perch-likes	Australia	Industrial	
3	Large pelagics (>=90 cm)	Perch-likes	Australia	Industrial	
4	Large pelagics (>=90 cm)	Perch-likes	Bahamas	Recreational	
5	Large pelagics (>=90 cm)	Perch-likes	Barbados	Recreational	
6	Large pelagics (>=90 cm)	Perch-likes	Bermuda (UK)	Artisanal	
	catch_type	reporting_status		gear_type	
1	Landings	Reported		longline	
2	Landings	Reported		longline	
3	Landings	Reported		longline	
4	Landings	Unreported	recreational fishing gear		
5	Landings	Unreported	recreational fishing gear		
6	Landings	Reported	small scale lines		
	end_use_type	tonnes	landed_value		
1	Fishmeal and fish oil	0.003373202	3.438170e-05		
2	Direct human consumption	6.739657104	1.040254e+04		
3	Other	0.003373202	3.438170e-05		
4	Direct human consumption	48.665105143	8.262728e+00		
5	Direct human consumption	0.009158559	4.846886e-04		
6	Direct human consumption	1.511085514	3.199403e+03		

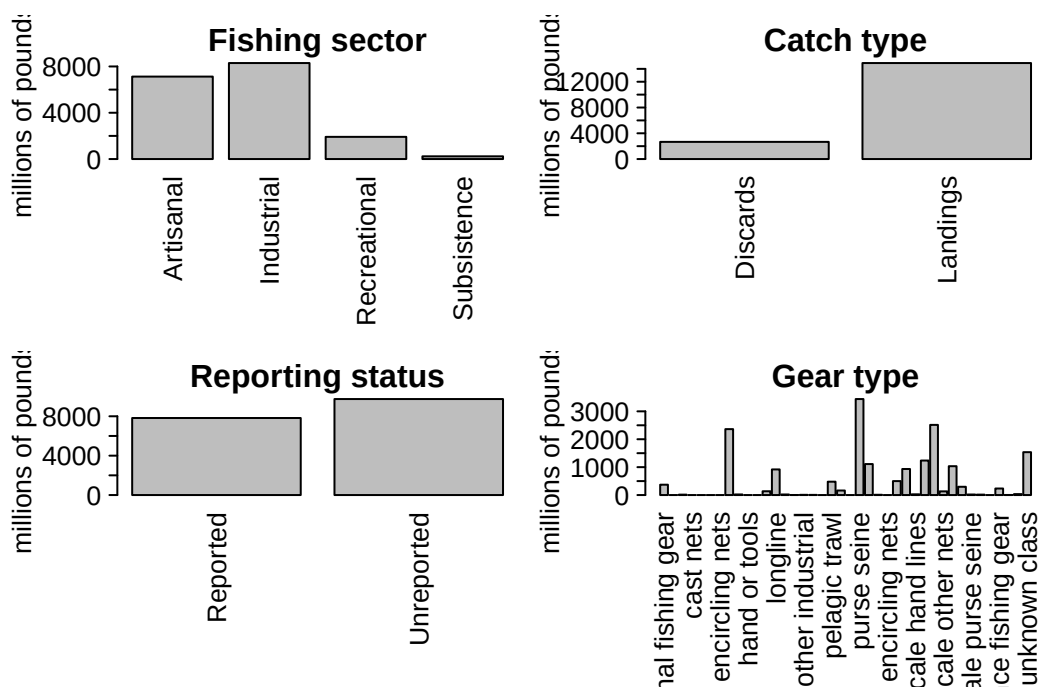
3 Analysis of international landings estimates from Sea Around Us database

```
#dim(d)
#apply(d[1:13], 2, table)

# convert tonnes to pounds
d$lbs <- d$tonnes * 1000 * 2.20462
```

We look at the characteristics of the landings within EEZs. These are largely industrial and artisanal in nature. Landings are the majority of the catch but discards are also present. There are more unreported landings than reported landings and a variety of gear types are used.

```
# look at characteristics of data
par(mfrow = c(2, 2), mar = c(12, 7, 3, 1), mex = 0.5, mgp = c(5, 1, 0))
barplot(tapply(d$lbs, d$fishing_sector, sum, na.rm = T)/10^6, las = 2, ylab = "millions of pounds")
barplot(tapply(d$lbs, d$catch_type, sum, na.rm = T)/10^6, las = 2, ylab = "millions of pounds")
barplot(tapply(d$lbs, d$reporting_status, sum, na.rm = T)/10^6, las = 2, ylab = "millions of pounds")
barplot(tapply(d$lbs, d$gear_type, sum, na.rm = T)/10^6, las = 2, ylab = "millions of pounds")
```



3.5 Separating the EEZs by area

Now we have to manually assign the different EEZs to respective areas. We separate the geographical areas within the Atlantic Ocean (North and South) and lump all other EEZs into “other oceans.” This latter category is largely the Pacific Ocean and to a lesser extent the Indian Ocean. We remove the high seas catch from this database as it is not ocean-specific and we have extracted the ocean-specific high seas catch previously.

```
# remove the high seas catch from this database
d <- d[which(d$area_type != "high_seas"), ]
d$fishing_entity <- as.factor(d$fishing_entity)
d$fishing_entity <- droplevels(d$fishing_entity)
table(d$area_type) # this should be all "eez"
```

```
eez
63056
```

```
# Western Central Atlantic EEZs
WCA <- c("Bahamas", "Barbados", "Cayman Isl. (UK)", "Dominica", "Dominican Republic", "Grenada",
        "Aruba (Netherlands)", "Haiti", "Guadeloupe (France)", "Jamaica", "Martinique (France)",
        "Montserrat (UK)", "Nicaragua (Caribbean)", "Puerto Rico (USA)", "Saint Kitts & Nevis",
        "Saint Lucia", "Saint Vincent & the Grenadines", "Trinidad & Tobago", "US Virgin Islands",
        "St Barthelemy (France)", "St Martin (France)", "Curaçao (Netherlands)", "Bonaire",
        "Saba and Sint Eustatius (Netherlands)", "Sint Maarten (Netherlands)", "Honduras",
        "Colombia (Caribbean)", "Mexico (Atlantic)", "Turks & Caicos Isl. (UK)", "Guyana",
        "Antigua & Barbuda", "Belize", "British Virgin Isl. (UK)", "French Guiana", "Cuba",
        "Anguilla (UK)", "Suriname", "Costa Rica (Caribbean)", "Guatemala (Caribbean)",
        "Panama (Caribbean)", "Bermuda (UK)", "Curaçao (Netherlands)")

# Brazil
Bra <- c("Brazil (mainland)", "Fernando de Noronha (Brazil)",
        "St Paul and St. Peter Archipelago (Brazil)")

# Canada
Can <- c("Canada (East Coast)", "Saint Pierre & Miquelon (France)")

# United States
USA <- c("USA (East Coast)", "USA (Gulf of Mexico)")
```


3 Analysis of international landings estimates from Sea Around Us database

```
# Northeast Atlantic EEZs
NEA <- c("Canary Isl. (Spain)", "Spain (mainland; Med and Gulf of Cadiz)", "Spain (Northwest)",
        "Spain (mainland, Med and Gulf of Cadiz)",
        "France (Atlantic Coast)", "Portugal (mainland)", "Azores Isl. (Portugal)")

# Eastern Central Atlantic EEZs
ECA <- c("Madeira Isl. (Portugal)", "Cape Verde", "Equatorial Guinea", "Benin",
        "Sao Tome & Principe ", "Sao Tome & Principe", "Morocco (Central)", "Ghana",
        "Côte d'Ivoire", "CÃ´te d'Ivoire", "Guinea", "Liberia", "Sierra Leone", "Nigeria",
        "Guinea-Bissau", "Senegal", "Cameroon", "Gambia", "Mauritania", "Morocco (South)",
        "Gabon", "Congo; R. of", "Congo, R. of", "Congo (ex-Zaire)")

# Southeastern Atlantic EEZs
SEA <- c("Angola", "Namibia", "South Africa (Atlantic and Cape)",
        "Ascension Isl. (UK)", "Saint Helena (UK)", "Tristan da Cunha Isl. (UK)")

# Southwest Atlantic EEZs
SWA <- c("Trindade & Martim Vaz Isl. (Brazil)", "Uruguay")

# Mediterranean EEZs
Med <- c("Italy (mainland)", "Libya", "Malta", "Syria", "Sicily (Italy)", "Sardinia (Italy)",
        "Balearic Islands (Spain)", "Cyprus (North)", "Cyprus (South)",
        "Gaza Strip", "Greece (without Crete)", "Crete (Greece)", "Lebanon",
        "Turkey (Mediterranean Sea)", "Egypt (Mediterranean)", "Israel (Mediterranean)",
        "Algeria", "Albania", "Croatia", "Montenegro", "Morocco (Mediterranean)",
        "France (Mediterranean)", "Corsica (France)")

# add variable to specify region
d$reg <- "other oceans"
d$reg[which(d$area_name %in% WCA)] <- "Western Central Atlantic EEZs"
d$reg[which(d$area_name %in% Bra)] <- "Brazil EEZ"
d$reg[which(d$area_name %in% Can)] <- "Canadian EEZ"
d$reg[which(d$area_name %in% USA)] <- "United States EEZ"
d$reg[which(d$area_name %in% NEA)] <- "Northeast Atlantic EEZs"
d$reg[which(d$area_name %in% ECA)] <- "Eastern Central Atlantic EEZs"
d$reg[which(d$area_name %in% SEA)] <- "Southeast Atlantic EEZs"
d$reg[which(d$area_name %in% SWA)] <- "Southwest Atlantic EEZs"
d$reg[which(d$area_name %in% Med)] <- "Mediterranean Sea EEZs"
```

Now we will check that we have classified the regions correctly.

3 Analysis of international landings estimates from Sea Around Us database

```
# check region designation
othlis <- unique(d$area_name[which(d$reg == "other oceans")])
print("EEZs of all other oceans: ")
```

```
[1] "EEZs of all other oceans: "
```

```
othlis
```

```
[1] "Australia"
[3] "Taiwan"
[5] "French Polynesia"
[7] "Andaman & Nicobar Isl. (India)"
[9] "Japan (main islands)"
[11] "Korea (South)"
[13] "Nauru"
[15] "Northern Marianas (USA)"
[17] "Marshall Isl."
[19] "Pakistan"
[21] "Philippines"
[23] "Viet Nam"
[25] "Tonga"
[27] "USA (West Coast)"
[29] "Wallis & Futuna Isl. (France)"
[31] "Guatemala (Pacific)"
[33] "Mexico (Pacific)"
[35] "Korea (North, Yellow Sea)"
[37] "Mauritius"
[39] "Christmas Isl. (Australia)"
[41] "Timor Leste"
[43] "Jarvis Isl. (USA)"
[45] "Indonesia (Central)"
[47] "Kiribati (Line Islands)"
[49] "New Caledonia (France)"
[51] "Lord Howe Isl. (Australia)"
[53] "Ecuador (mainland)"
[55] "Fiji"
[57] "Norfolk Isl. (Australia)"
[59] "Pitcairn (UK)"
[61] "Costa Rica (Pacific)"
"China"
"Mayotte (France)"
"Hong Kong (China)"
"Indonesia (Eastern)"
"Japan (Daito Islands)"
"Hawaii Northwest Islands (USA)"
"Niue (New Zealand)"
"Micronesia (Federated States of)"
"Palau"
"Papua New Guinea"
"Seychelles"
"Tokelau (New Zealand)"
"Hawaii Main Islands (USA)"
"Wake Isl. (USA)"
"Samoa"
"Kiribati (Gilbert Islands)"
"Japan (Ogasawara Islands)"
"Korea (North, Sea of Japan)"
"Solomon Isl."
"Johnston Atoll (USA)"
"Palmyra Atoll & Kingman Reef (USA)"
"Howland & Baker Isl. (USA)"
"Indonesia (Indian Ocean)"
"Kiribati (Phoenix Islands)"
"Vanuatu"
"Myanmar"
"El Salvador"
"Malaysia (Peninsula West)"
"Peru"
"Colombia (Pacific)"
"Nicaragua (Pacific)"
```

3 Analysis of international landings estimates from Sea Around Us database

[63] "Panama (Pacific)"	"Thailand (Andaman Sea)"
[65] "Thailand (Gulf of Thailand)"	"Cook Islands"
[67] "Kermadec Isl. (New Zealand)"	"Tuvalu"
[69] "Honduras (Pacific)"	"American Samoa"
[71] "Galapagos Isl. (Ecuador)"	"New Zealand"
[73] "Brunei Darussalam"	"Malaysia (Sabah)"
[75] "Malaysia (Sarawak)"	"Guam (USA)"
[77] "Clipperton Isl. (France)"	"South Africa (Indian Ocean Coast)"
[79] "Singapore"	"Easter Isl. (Chile)"
[81] "Madagascar"	"Réunion (France)"
[83] "Chagos Archipelago (UK)"	"Comoros Isl."
[85] "Mozambique Channel Isl. (France)"	"Kenya"
[87] "Mozambique"	"Somalia"
[89] "Tanzania"	"Glorieuse Islands (France)"
[91] "Yemen (Socotra)"	"Oman"
[93] "St Paul & Amsterdam Isl. (France)"	"Iran (Sea of Oman)"
[95] "Desventuradas Isl. (Chile)"	"India (mainland)"
[97] "Maldives"	"Tromelin Isl. (France)"
[99] "Cambodia"	"Sri Lanka"
[101] "Malaysia (Peninsula East)"	"Bangladesh"

```
print("U.S. EEZs outside the Atlantic Ocean: ")
```

```
[1] "U.S. EEZs outside the Atlantic Ocean: "
```

```
othlis[grepl("USA", othlis)]
```

[1] "Hawaii Northwest Islands (USA)"	"Northern Marianas (USA)"
[3] "Hawaii Main Islands (USA)"	"USA (West Coast)"
[5] "Wake Isl. (USA)"	"Johnston Atoll (USA)"
[7] "Palmyra Atoll & Kingman Reef (USA)"	"Jarvis Isl. (USA)"
[9] "Howland & Baker Isl. (USA)"	"Guam (USA)"

```
print("French EEZs outside the Atlantic Ocean: ")
```

```
[1] "French EEZs outside the Atlantic Ocean: "
```

3 Analysis of international landings estimates from Sea Around Us database

```
othlis[grep("France", othlis)]
```

```
[1] "Mayotte (France)"           "Wallis & Futuna Isl. (France)"
[3] "New Caledonia (France)"     "Clipperton Isl. (France)"
[5] "Réunion (France)"          "Mozambique Channel Isl. (France)"
[7] "Glorieuse Islands (France)" "St Paul & Amsterdam Isl. (France)"
[9] "Tromelin Isl. (France)"
```

```
unique(d$area_name[which(d$reg == "Western Central Atlantic EEZs")]) # Western Central EE
```

```
[1] "Bahamas"
[2] "Barbados"
[3] "Bermuda (UK)"
[4] "Cayman Isl. (UK)"
[5] "Dominica"
[6] "Dominican Republic"
[7] "Grenada"
[8] "Guadeloupe (France)"
[9] "Haiti"
[10] "Jamaica"
[11] "Martinique (France)"
[12] "Montserrat (UK)"
[13] "Puerto Rico (USA)"
[14] "Saint Kitts & Nevis"
[15] "Saint Lucia"
[16] "Saint Vincent & the Grenadines"
[17] "Trinidad & Tobago"
[18] "US Virgin Isl."
[19] "St Barthelemy (France)"
[20] "St Martin (France)"
[21] "Curaçao (Netherlands)"
[22] "Bonaire (Netherlands)"
[23] "Saba and Sint Eustatius (Netherlands)"
[24] "Sint Maarten (Netherlands)"
[25] "Honduras (Caribbean)"
[26] "Colombia (Caribbean)"
[27] "Mexico (Atlantic)"
[28] "Turks & Caicos Isl. (UK)"
[29] "Costa Rica (Caribbean)"
```

3 Analysis of international landings estimates from Sea Around Us database

```
[30] "Panama (Caribbean)"
[31] "Antigua & Barbuda"
[32] "Belize"
[33] "British Virgin Isl. (UK)"
[34] "Cuba"
[35] "French Guiana"
[36] "Guyana"
[37] "Anguilla (UK)"
[38] "Suriname"
[39] "Venezuela"
[40] "Guatemala (Caribbean)"
[41] "Aruba (Netherlands)"
[42] "Nicaragua (Caribbean)"
```

```
unique(d$area_name[which(d$reg == "Brazil EEZ")]) # Brazil EEZs:
```

```
[1] "Brazil (mainland)"
[2] "Fernando de Noronha (Brazil)"
[3] "St Paul and St. Peter Archipelago (Brazil)"
```

```
unique(d$area_name[which(d$reg == "Canadian EEZ")]) # Canadian EEZs:
```

```
[1] "Canada (East Coast)" "Saint Pierre & Miquelon (France)"
```

```
unique(d$area_name[which(d$reg == "United States EEZ")]) # United States EEZs:
```

```
[1] "USA (East Coast)" "USA (Gulf of Mexico)"
```

```
unique(d$area_name[which(d$reg == "Northeast Atlantic EEZs")]) # Northeast Atlantic EEZs:
```

```
[1] "Azores Isl. (Portugal)"
[2] "Spain (mainland, Med and Gulf of Cadiz)"
[3] "Canary Isl. (Spain)"
[4] "Portugal (mainland)"
[5] "France (Atlantic Coast)"
[6] "Spain (Northwest)"
```

3 Analysis of international landings estimates from Sea Around Us database

```
unique(d$area_name[which(d$reg == "Eastern Central Atlantic EEZs")]) # Eastern Central At
```

[1] "Cape Verde"	"Equatorial Guinea"
[3] "Sao Tome & Principe"	"Madeira Isl. (Portugal)"
[5] "Togo"	"Ghana"
[7] "Guinea"	"Liberia"
[9] "Guinea-Bissau"	"Senegal"
[11] "Sierra Leone"	"Morocco (Central)"
[13] "Morocco (South)"	"Côte d'Ivoire"
[15] "Gabon"	"Benin"
[17] "Gambia"	"Mauritania"
[19] "Nigeria"	"Cameroon"
[21] "Congo, R. of"	"Congo (ex-Zaire)"

```
unique(d$area_name[which(d$reg == "Southeast Atlantic EEZs")]) # Southeast Atlantic EEZs:
```

[1] "Ascension Isl. (UK)"	"South Africa (Atlantic and Cape)"
[3] "Saint Helena (UK)"	"Angola"
[5] "Namibia"	"Tristan da Cunha Isl. (UK)"

```
unique(d$area_name[which(d$reg == "Southwest Atlantic EEZs")]) # Southwest Atlantic EEZs:
```

[1] "Trindade & Martim Vaz Isl. (Brazil)"	"Uruguay"
---	-----------

```
unique(d$area_name[which(d$reg == "Mediterranean Sea EEZs")]) # Mediterranean Sea EEZs:
```

[1] "Italy (mainland)"	"Libya"
[3] "Malta"	"Syria"
[5] "Sicily (Italy)"	"Sardinia (Italy)"
[7] "Balearic Islands (Spain)"	"Tunisia"
[9] "Algeria"	"Morocco (Mediterranean)"
[11] "France (Mediterranean)"	"Corsica (France)"
[13] "Croatia"	"Lebanon"
[15] "Albania"	"Greece (without Crete)"
[17] "Crete (Greece)"	"Cyprus (South)"

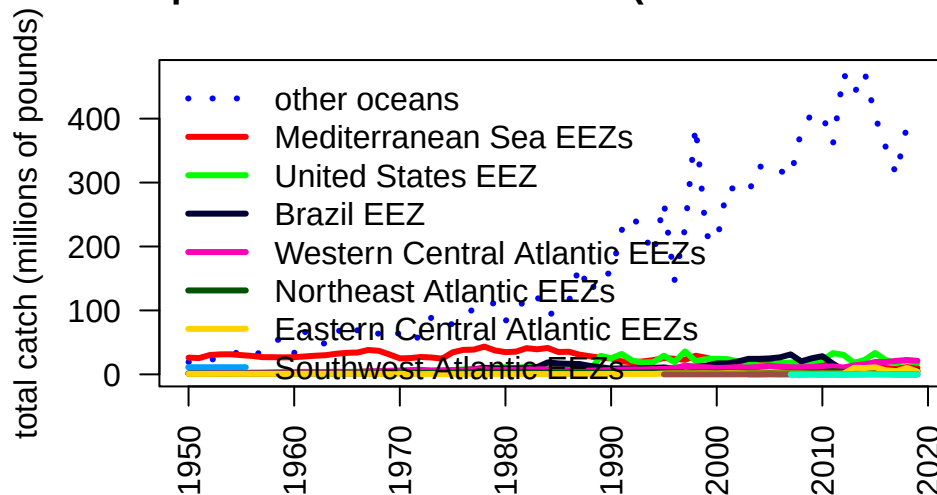
3.6 Summarize catch by regional EEZs

Now we will summarize the catch by region by grouping the catch across the EEZs by region and plot the results. The U.S. landings should be very close to what is reported in Section 2 because the majority of U.S. catch is reported through the various data collection systems that are compiled within Sea Around Us.

```
# summarize EEZ catch by region
tab <- tapply(d$lbs, list(d$year, d$reg), sum, na.rm = T)
#head(tab)
tab2 <- t(tab)
tab3 <- tab2[order(rowSums(tab2, na.rm = T), decreasing = T), ]
tab_glob <- t(tab3)
#head(tab_glob)
yrs <- as.numeric(rownames(tab))

par(mar = c(5, 4, 4, 3), mgp = c(3, 1, 0))
matplot(yrs, tab_glob/10^6, las = 2, type = "l", lty = c(3, rep(1, 9)), lwd = 3, col = glasbey(ncol(tab_glob)),
        xlab = "", ylab = "total catch (millions of pounds)",
        main = "Global dolphin catches within EEZs (commercial + recreational)")
legend("topleft", colnames(tab_glob), col = glasbey(ncol(tab_glob)),
      lty = c(3, rep(1, 9)), lwd = 3, bty = "n")
```

Global dolphin catches within EEZs (commercial + recreation)

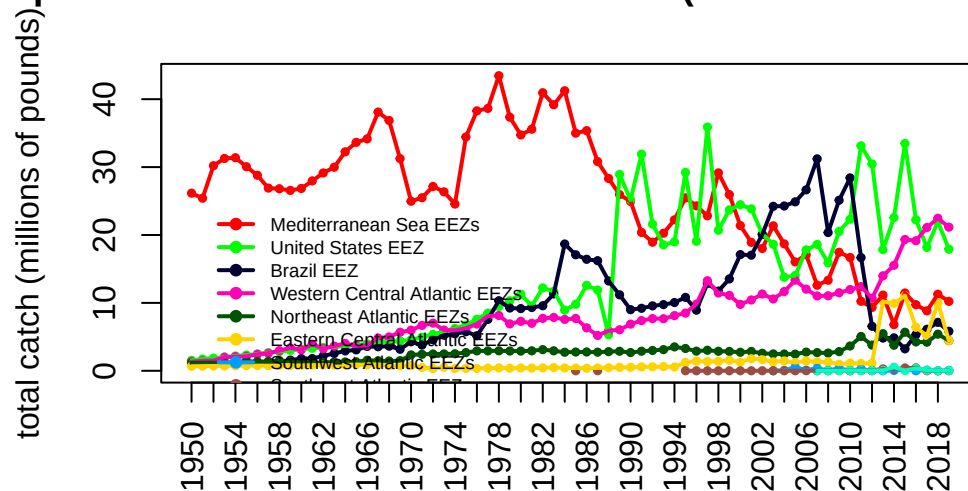


We can see that the EEZ catches globally are dominated by areas outside of the Atlantic ("other oceans") so we will replot the data after moving this category. Here we can see that most of the catches historically were coming from the Mediterranean Sea, though these catches have been declining drastically since the 1970s. Catches in the United States have showed an opposite pattern, with low historical catches that increased in the 1980s (largely due to the introduction of recreational monitoring). Catches in the Western Central EEZs have increased gradually over time.

```
# remove other oceans
#head(tab_glob)
tab <- tab_glob[, -which(colnames(tab_glob) == "other oceans")]
#head(tab)

matplot(yrs, tab/10^6, las = 2, type = "b", pch = 19, cex = 0.5, lty = 1, col = glasbey(ncol(tab)+1)[-1],
        xlab = "", ylab = "total catch (millions of pounds)", axes = F,
        main = "Dolphin catches from the Atlantic EEZs (commercial + recreational)")
matplot(yrs, tab/10^6, las = 2, type = "l", pch = 19, lty = 1, lwd = 2, add = T, col = glasbey(ncol(tab)+1)[-1])
axis(1, at = yrs[seq(1, 90, 2)], las = 2); axis(2); box()
legend(1952, 25, colnames(tab), col = glasbey(ncol(tab)+1)[-1], pch = 19, lty = 1, lwd = 2,
```


olphin catches from the Atlantic EEZs (commercial + recrea



3.7 Merge high seas with catch with exclusive economic zones

Now we will merge in the high seas catch from the steps above into the EEZ catches to get a better picture of total catch in FAO zones 31 and 21. Recall that the U.S. Gulf and Atlantic catches are not necessarily parsed apart correctly in the public databases which get compiled into Sea Around Us, so these will have the same deficiencies as noted in Section 2. Later we will merge in the corrected data.

```
hs <- read.csv("data/SAU/high_seas_by_year.csv")
names(hs)[1] <- "year"

# check that columns match
#head(tab)
#head(hs)
table(rownames(tab) == hs$year)
```

TRUE

70

3 Analysis of international landings estimates from Sea Around Us database

```
tab[which(is.na(tab))] <- 0
tab_atl <- cbind(tab, hs[2:3])

# check the merge
head(tab_atl)
```

	Mediterranean Sea EEZs	United States EEZ	Brazil EEZ
1950	26162262	1548245	1127242
1951	25410830	1721403	1060734
1952	30206350	1871779	1128992
1953	31275619	2024535	1111008
1954	31383840	2177865	1090492
1955	30056210	2309815	1190424

	Western Central Atlantic EEZs	Northeast Atlantic EEZs
1950	1345027	1173441
1951	1467925	1187386
1952	1510406	1193045
1953	2001465	1141504
1954	2069754	1269523
1955	2130341	1405543

	Eastern Central Atlantic EEZs	Southwest Atlantic EEZs
1950	677552.5	0
1951	693004.4	0
1952	708466.7	0
1953	723939.4	0
1954	739422.5	0
1955	754915.9	0

	Southeast Atlantic EEZs	Canadian EEZ	Atlantic..Northwest
1950	0	0	0
1951	0	0	0
1952	0	0	0
1953	0	0	0
1954	0	0	0
1955	0	0	0

	Atlantic..Western.Central
1950	0
1951	0
1952	0
1953	0
1954	0

3 Analysis of international landings estimates from Sea Around Us database

1955

0

```
# separate all WCA catches
unique(d$reg) # use Western Central, U.S. and Canada

[1] "other oceans"           "Western Central Atlantic EEZs"
[3] "Brazil EEZ"             "Eastern Central Atlantic EEZs"
[5] "Mediterranean Sea EEZs" "Northeast Atlantic EEZs"
[7] "United States EEZ"      "Southeast Atlantic EEZs"
[9] "Southwest Atlantic EEZs" "Canadian EEZ"

dwc <- d[which(d$reg == "Western Central Atlantic EEZs" | d$reg == "United States EEZ" | d$reg == "Canadian EEZ"), ]

#unique(dwc$area_name[which(dwc$reg == "United States EEZ")])
#unique(dwc$area_name[which(dwc$reg == "Canadian EEZ")])
#unique(dwc$area_name[which(dwc$reg == "Western Central Atlantic EEZs")])

# relabel U.S. by area
dwc$reg[which(dwc$area_name == "USA (East Coast)")] <- "United States East Coast"
dwc$reg[which(dwc$area_name == "USA (Gulf of Mexico)")] <- "United States Gulf"
dwc$reg[which(dwc$area_name == "Puerto Rico (USA)")] <- "United States Caribbean"
dwc$reg[which(dwc$area_name == "US Virgin Isl.")] <- "United States Caribbean"
#table(dwc$area_name, dwc$reg)

tab_wc <- tapply(dwc$lbs, list(dwc$year, dwc$reg), sum, na.rm = T)
#head(tab_wc)

# reorder S to N
tab_wc <- tab_wc[, c(5, 2, 4, 3, 1)]
head(tab_wc)
```

	Western Central Atlantic EEZs	United States Caribbean	United States Gulf
1950	1202464	142562.8	309225.3
1951	1299559	168365.6	339921.3
1952	1341943	168463.5	370512.4
1953	1839330	162135.2	400578.5
1954	1901095	168659.3	433479.6
1955	1954511	175830.1	463752.0
	United States East Coast	Canadian EEZ	

3 Analysis of international landings estimates from Sea Around Us database

1950	1239020	NA
1951	1381482	NA
1952	1501267	NA
1953	1623956	NA
1954	1744386	NA
1955	1846063	NA

```
# merge in the high seas catch - check years match
#head(hs)
table(rownames(tab_wc) == hs$year)
```

```
TRUE
70
```

```
high_seas <- hs$Atlantic..Western.Central + hs$Atlantic..Northwest
high_seas[high_seas == 0] <- NA
tab_wc <- cbind(tab_wc, high_seas)
head(tab_wc)
```

	Western Central Atlantic EEZs	United States	Caribbean	United States Gulf
1950	1202464		142562.8	309225.3
1951	1299559		168365.6	339921.3
1952	1341943		168463.5	370512.4
1953	1839330		162135.2	400578.5
1954	1901095		168659.3	433479.6
1955	1954511		175830.1	463752.0

	United States East Coast	Canadian EEZ	high_seas
1950	1239020	NA	NA
1951	1381482	NA	NA
1952	1501267	NA	NA
1953	1623956	NA	NA
1954	1744386	NA	NA
1955	1846063	NA	NA

3.8 Plot Western Atlantic catches

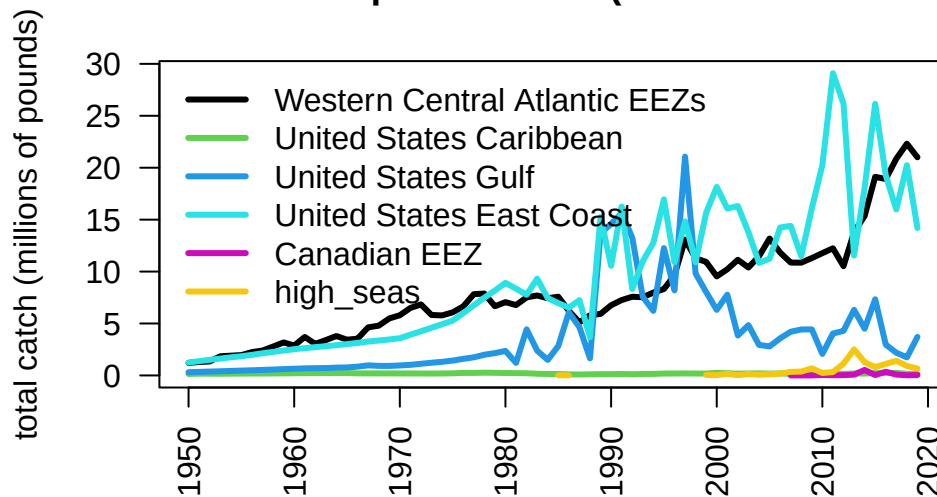
Now we plot the total Western Atlantic catches – both within the EEZs and the high seas – separating by U.S., Canada, and the Western Central Atlantic (i.e., the Wider

3 Analysis of international landings estimates from Sea Around Us database

Caribbean). Most of the catch is coming from the United States and the 40 combined Caribbean jurisdictions which comprise the Western Central Atlantic.

```
par(mar = c(5, 4, 4, 3), mgp = c(3, 1, 0))
matplot(yrs, tab_wc/10^6, las = 2, type = "l", lty = 1, lwd = 3, col = c(1, 3:7),
        xlab = "", ylab = "total catch (millions of pounds)",
        main = "Western Atlantic dolphin catches (commercial + recreational)")
legend("topleft", colnames(tab_wc), col = c(1, 3:7), lty = 1, lwd = 3, bty = "n")
```

Western Atlantic dolphin catches (commercial + recreational)



Separating out the mainland United States catches, here are the international and domestic (mainland) catches plotted over time.

```
tab_int <- tab_wc[, c(1, 2, 5, 6)]
head(tab_int)
```

	Western Central Atlantic EEZs	United States Caribbean	Canadian EEZ
1950	1202464	142562.8	NA
1951	1299559	168365.6	NA
1952	1341943	168463.5	NA
1953	1839330	162135.2	NA
1954	1901095	168659.3	NA

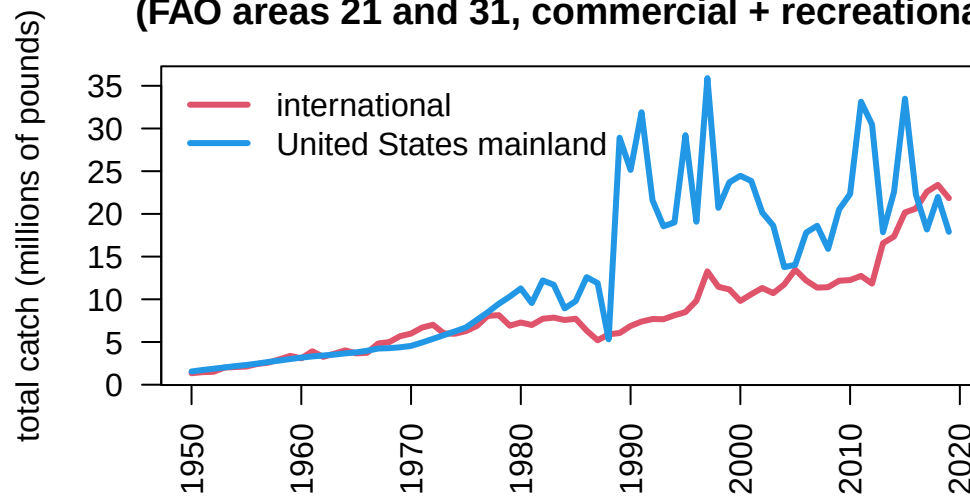
3 Analysis of international landings estimates from Sea Around Us database

1955		1954511	175830.1	NA
	high_seas			
1950	NA			
1951	NA			
1952	NA			
1953	NA			
1954	NA			
1955	NA			

```
int_catch <- round(rowSums(tab_int, na.rm = T), 2)
dom_catch <- rowSums(tab_wc[, c(3, 4)], na.rm = T)

matplot(yrs, cbind(int_catch, dom_catch)/10^6, las = 2, type = "l", lty = 1, lwd = 3, col =
        xlab = "", ylab = "total catch (millions of pounds)",
        main = "Western Atlantic dolphin catches in international vs domestic waters\n(FAO
legend("topleft", c("international", "United States mainland"), col = c(2, 4), lty = 1, lwd
```

Western Atlantic dolphin catches in international vs domestic (FAO areas 21 and 31, commercial + recreational)



```
# output international catches
write.csv(data.frame(int_catch), file = "data/international_catches.csv", row.names = T)
```

3.9 Look at Venezuela catches in detail

Note in the plot above a rapid increase in international catches within the Western Atlantic. This is driven by a sudden increase in catches reported within Sea Around Us within the EEZ. We investigate these catches in detail as it seems odd that a relatively small fleet would be able to ramp up effort so rapidly. Data can be downloaded on a country-specific basis, the catches by taxon in the waters of Venezuela are accessed [here](#).

```
rm(list = ls()) # clear the workspace

# download the data for waters of Venezuela
ven <- read.csv("data/SAU/SAU EEZ 862 v50-1.csv")

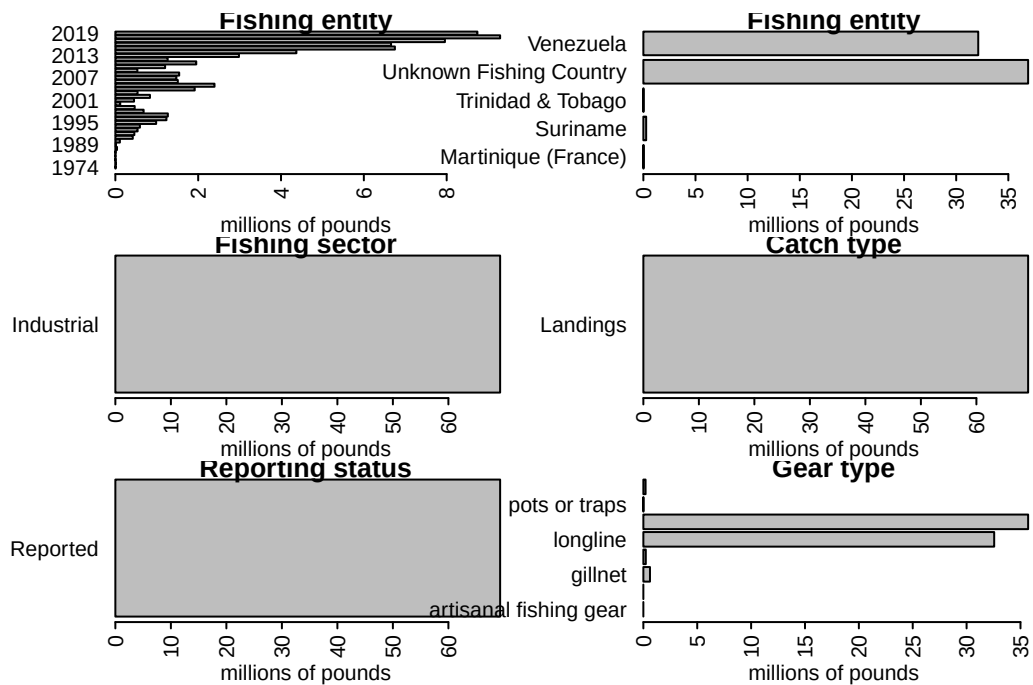
# subset dolphinfish from the data
v <- ven[which(ven$common_name == "Common dolphinfish"), ]

#apply(v[1:13], 2, table)

#convert to pounds
v$pounds <- v$tonnes * 1000 * 2.20462

# look at the characteristics of the data
par(mar = c(5, 10, 1, 1), mfrow = c(3, 2), mex = 0.5)
#barplot(tapply(v$pounds/10^6, v$area_name, sum, na.rm = T), las = 2, xlab = "millions of p
#barplot(tapply(v$pounds/10^6, v$area_type, sum, na.rm = T), las = 2, xlab = "millions of p
barplot(tapply(v$pounds/10^6, v$year, sum, na.rm = T), las = 2, xlab = "millions of pounds"
barplot(tapply(v$pounds/10^6, v$fishing_entity, sum, na.rm = T), las = 2, xlab = "millions
barplot(tapply(v$pounds/10^6, v$fishing_sector, sum, na.rm = T), las = 2, xlab = "millions
barplot(tapply(v$pounds/10^6, v$catch_type, sum, na.rm = T), las = 2, xlab = "millions of p
barplot(tapply(v$pounds/10^6, v$reporting_status, sum, na.rm = T), las = 2, xlab = "million
barplot(tapply(v$pounds/10^6, v$gear_type, sum, na.rm = T), las = 2, xlab = "millions of po
```

3 Analysis of international landings estimates from Sea Around Us database



Note that a large portion of the catch in the database is listed as coming from an “Unknown Fishing Country.” We extract these catches and further investigate the characteristics.

```
# extract the Unknown Fishing Country catch
v1 <- v[which(v$fishing_entity == "Unknown Fishing Country"), ]
apply(v1[1:15], 2, table)
```

\$area_name

Venezuela
39

\$area_type

eez
39

\$data_layer

Assigned Tuna RFMO catch

3 Analysis of international landings estimates from Sea Around Us database

39

\$uncertainty_score
< table of extent 0 >

\$year

2013	2014	2015	2016	2017	2018	2019
5	5	5	6	7	5	6

\$scientific_name

Coryphaena hippurus
39

\$common_name

Common dolphinfish
39

\$functional_group

Large pelagics (≥ 90 cm)
39

\$commercial_group

Perch-likes
39

\$fishing_entity

Unknown Fishing Country
39

\$fishing_sector

Industrial
39

\$catch_type

3 Analysis of international landings estimates from Sea Around Us database

Landings

39

\$reporting_status

Reported

39

\$gear_type

artisanal fishing gear	bottom trawl	gillnet
1	5	7
hand lines	longline	pole and line
3	4	7
pots or traps	unknown by source	
5	7	

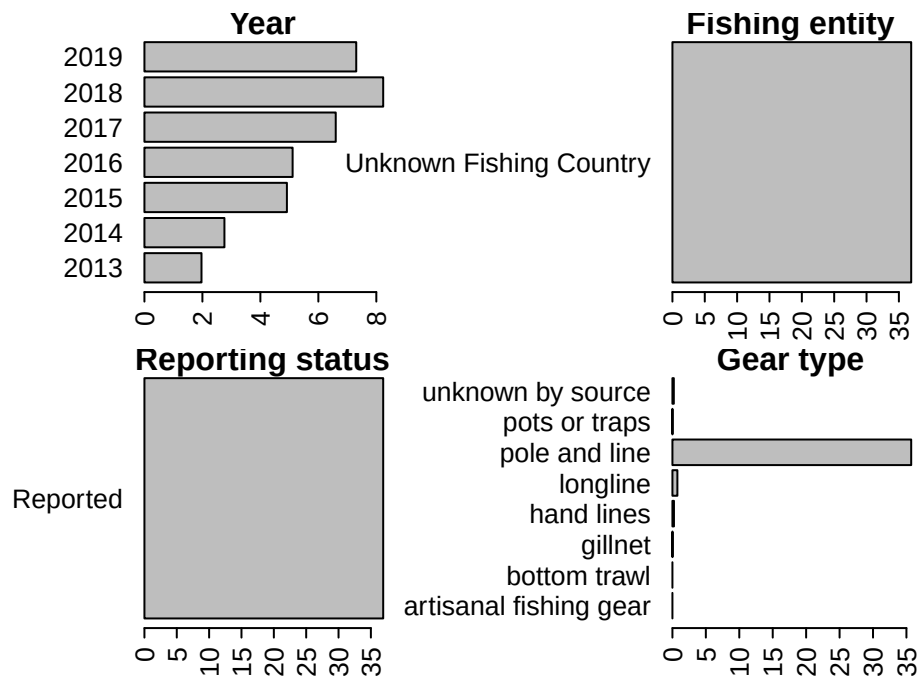
\$end_use_type

Direct human consumption

39

```
# look at characteristics
par(mar = c(3, 10, 1, 5), mfrow = c(2, 2), mex = 0.6)
barplot(tapply(v1$pounds/10^6, v1$year, sum, na.rm = T), las = 2, xlab = "millions of pound
barplot(tapply(v1$pounds/10^6, v1$fishing_entity, sum, na.rm = T), las = 2, xlab = "million
barplot(tapply(v1$pounds/10^6, v1$reporting_status, sum, na.rm = T), las = 2, xlab = "milli
barplot(tapply(v1$pounds/10^6, v1$gear_type, sum, na.rm = T), las = 2, xlab = "millions of
```

3 Analysis of international landings estimates from Sea Around Us database



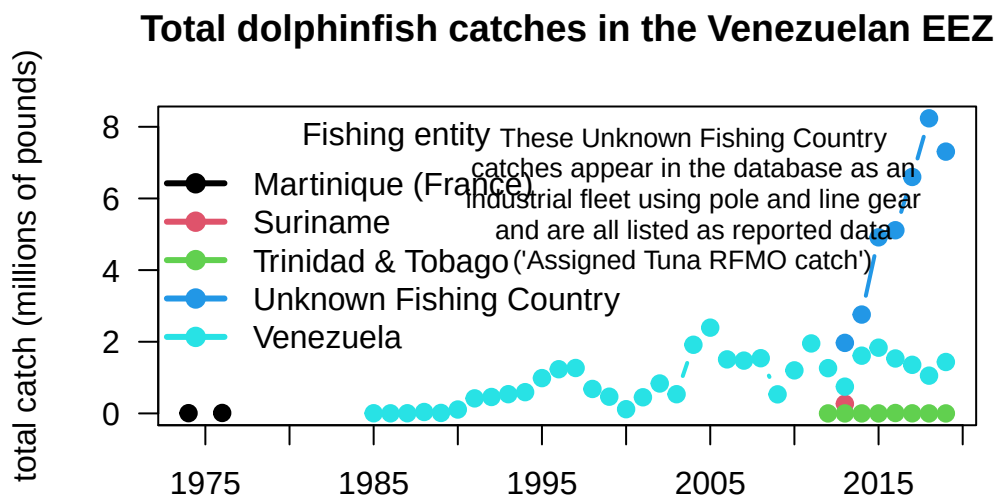
Here is another look at the catches within the venezuelan EEZ, separated by fishing entity.

```
# summarize Venezuela EEZ catches by fishing entity
tab <- tapply(v$pounds, list(v$year, v$fishing_entity), sum, na.rm = T)
head(tab)
```

	Martinique (France)	Suriname	Trinidad & Tobago	Unknown Fishing Country
1974	7320.429	NA	NA	NA
1976	12814.641	NA	NA	NA
1985	NA	NA	NA	NA
1986	NA	NA	NA	NA
1987	NA	NA	NA	NA
1988	NA	NA	NA	NA
Venezuela				
1974	NA			
1976	NA			
1985	4699.563			
1986	2229.484			
1987	4585.610			
1988	39960.149			

3 Analysis of international landings estimates from Sea Around Us database

```
matplot(rownames(tab), tab/10^6, las = 2, type = "b", lty = 1, lwd = 2, pch = 19, col = 1:5,
        xlab = "", ylab = "total catch (millions of pounds)", axes = F,
        main = "Total dolphinfish catches in the Venezuelan EEZ")
axis(1, at = seq(1950, 2020, 5)); axis(2, las = 2); box()
legend("topleft", colnames(tab), col = 1:5, pch = 19, lty = 1, lwd = 3, bty = "n", title =
text(2004, 6, cex = 0.8,
     "These Unknown Fishing Country\ncatches appear in the database as an\nindustrial fleet
```



The very substantial increase in international Western Atlantic dolphin catches can be attributed solely to this Unknown Fishing Country catches which appear in the database beginning in 2013. The documentation of the catch reconstruction can be found in the Sea Around Us documentation files (see Mendoza et al. 2015). We spoke to Jeremy Mendoza and other experts who were involved in the reconstruction of domestic catches in Venezuela, and they noted that the "Assigned tuna RFMO catch" actually emerge from spatially reshuffled landing reports to ICCAT, according to the methods of Coulter et al. (2020). In this work, the Sea Around Us team developed a global spatial reallocation of landings of tunas and associated species that were reported to ICCAT. In short, this work uses the fraction of reported spatially disaggregated data to reallocate all data reported to ICCAT. This step is necessary because there are not many countries that report spatial data to ICCAT in the region, and this method may

3 Analysis of international landings estimates from Sea Around Us database

lead to a significant reallocation of landings from the Caribbean to the Venezuelan EEZ, where the local pole and line fishery is significant (J. Mendoza, personal communication). Thus, it is likely that landings from other EEZs, that were included in the reconstructed catch data layer, could have been reassigned to the Venezuelan EEZ.

Because there may be some overlap between the reconstructed data used in the national EEZ estimates, and the ICCAT reporting that was reassigned, there could potentially be a “double counting” of these catch estimates in the Sea Around Us database. These Unknown Fishing Country catches that lead to a large increase in total estimated catches within the Venezuelan EEZ should be treated with caution at this time, until we can verify with Sea Around Us database managers that the spatial disaggregation method used has not led to an overestimate.

3.10 References

Pauly D., Zeller D., Palomares M.L.D. (Editors), 2020. Sea Around Us Concepts, Design and Data (seararoundus.org).

4 NOAA Fisheries landings data

Here we will explore commercial and recreational landings data accessed through the FOSS (Fisheries One Stop Shop) data portal. The advantage of this portal is that commercial and recreational data are available side-by-side in the same units (pounds landed).

4.1 Data access and upload

Data are downloaded from the NOAA Fisheries FOSS site. For the Data Set, we choose both commercial and recreational. For the Years, we select all years from 1980 to the present. For NMFS Region, we select the "NMFS Regions" button and then select the Gulf, South Atlantic, Middle Atlantic and New England from the box. For the Species, we select "Dolphinfish" and "Dolphinfish**." For Report Format, we choose TOTALS BY YEAR/REGION. Hitting the "RUN REPORT" button then generates the data needed for the subsequent analysis, which can be downloaded as a .csv file under the "Actions" button.

We then read in the data file and clean it for analysis (removing the commas from the numbers so that R recognizes the values as numbers). We output the file header to make sure it was read in and that the numbers were converted correctly.

```
# clear workspace
rm(list = ls())

# read in data file
d <- read.table("data/FOSS_landings.csv", header = T, sep = ",", skip = 1)
head(d)
```

	Year	Region.Name	Pounds	Dollars	Collection	Metric.Tons
1	1980	Gulf	94,984	55,676	Commercial	43
2	1980	Middle Atlantic	400	256	Commercial	0
3	1980	New England	200	54	Commercial	0

4 NOAA Fisheries landings data

4	1980	South Atlantic	76,042	45,852	Commercial	34
5	1981	Gulf	73,075	46,951	Commercial	33
6	1981	Gulf	1,559,544		Recreational	707

```
# take commas out of file -----
d$Pounds <- as.numeric(gsub(",", "", d$Pounds))
d$Dollars <- as.numeric(gsub(",", "", d$Dollars))
d$Metric.Tons <- as.numeric(gsub(",", "", d$Metric.Tons))
head(d)
```

	Year	Region.Name	Pounds	Dollars	Collection	Metric.Tons
1	1980	Gulf	94984	55676	Commercial	43
2	1980	Middle Atlantic	400	256	Commercial	0
3	1980	New England	200	54	Commercial	0
4	1980	South Atlantic	76042	45852	Commercial	34
5	1981	Gulf	73075	46951	Commercial	33
6	1981	Gulf	1559544	NA	Recreational	707

4.2 Summarize the landings

We then summarize the landings (in pounds) by year, sector, and region. The result is a new data frame with each row representing a year and each column representing the total landings for each region and sector.

```
# compile by year for sectors and regions -----
d$Collection <- factor(d$Collection, levels = c("Recreational", "Commercial"))
tab <- tapply(d$Pounds, list(d$Year, d$Collection, d$Region.Name), sum, na.rm = T)
tab[is.na(tab)] <- 0
#head(tab)

dat <- data.frame(cbind(tab[, , "Gulf"], tab[, , "South Atlantic"],
                        tab[, , "Middle Atlantic"], tab[, , "New England"]))
names(dat) <- c("rec_Gulf", "com_Gulf", "rec_SA", "com_SA", "rec_MA", "com_MA", "rec_NE", "com_NE")
dat
```

	rec_Gulf	com_Gulf	rec_SA	com_SA	rec_MA	com_MA	rec_NE	com_NE
1980	0	94984	0	76042	0	400	0	200
1981	1559544	73075	8421898	59037	0	0	0	0

4 NOAA Fisheries landings data

1982	4197503	162457	7573035	143960	2097	200	0	0
1983	2162936	190808	9134022	127986	45927	1400	0	400
1984	1189573	282976	7265430	158901	0	1700	0	400
1985	2542089	269477	6684821	142684	113666	5000	0	4800
1986	5540003	492212	6190187	189747	125653	4200	0	200
1987	4209638	403703	6883546	212905	121433	13400	0	1100
1988	1014312	491610	3134070	212709	130240	26600	2196	17800
1989	12551461	1029628	13273504	426392	1386752	81700	437	15300
1990	13389540	1098819	9283973	514980	624928	69106	2714	14233
1991	13760600	1674124	14756442	650333	722090	90722	6294	9816
1992	12481422	714369	5893700	326831	860112	74423	1411	10315
1993	6873820	596967	7077105	520512	3001335	99420	200927	27069
1994	5448709	679903	10554836	618614	1400635	101044	12586	12328
1995	11142952	1177325	14318412	1164608	1096473	198425	24061	13811
1996	7086006	1047020	9128895	548682	1077364	43581	0	7788
1997	19958088	989648	12284366	877303	1422057	105559	0	13265
1998	9324186	422501	7312090	406368	2920384	86502	120322	11553
1999	7329953	612445	13973364	504263	1005852	98927	4458	5990
2000	5641491	582576	16049049	537077	1433991	36048	0	6545
2001	7355385	368503	14945284	470897	477081	51342	5189	29134
2002	3513226	290601	13794853	436472	1295470	83534	600948	14836
2003	4481937	310836	12112718	459351	929763	46851	129513	21275
2004	2402315	435341	9707678	555973	427145	31916	0	29042
2005	2537132	207695	10299670	376989	437472	19703	0	19982
2006	3290474	224668	12805952	433121	887684	27724	0	19645
2007	3754513	370397	13164377	740255	241212	79832	19542	44707
2008	3937153	384954	10417536	627939	264017	57764	0	16328
2009	3860527	496701	13251102	1007355	1538112	55271	0	92010
2010	1216198	111790	8543542	572031	813738	35869	0	14939
2011	3023914	309324	10774812	543811	2173136	15376	187289	26587
2012	3229090	146256	9983276	704240	755558	44918	26892	46171
2013	5425636	187966	8483569	583556	302421	41936	226988	18564
2014	3074373	201943	11925345	1041856	2129222	58868	183673	61505
2015	6286577	154133	14821772	920454	4501793	36190	1198628	37808
2016	1777348	37411	12357817	828133	2212513	61336	16499	15747
2017	2064106	65141	9457460	532973	1820541	16179	48607	30969
2018	1626095	56188	11713866	435059	3104018	20514	97107	12667
2019	3567170	83956	7967978	563132	1518629	48224	66	16918
2020	1952241	26332	5653294	260311	679556	14040	0	1880
2021	2893817	26570	6061024	178822	1051785	25318	123038	11421
2022	1664771	15906	6845573	182798	708494	17360	466897	9131

4 NOAA Fisheries landings data

2023 1427518 13305 6630236 125302 734866 28467 122550 8310

4.3 Plot the summary

Now we plot the data, using similar color schemes by region. Because the recreational data have high imprecision prior to the 1990s we plot from 1990 to the last year of data.

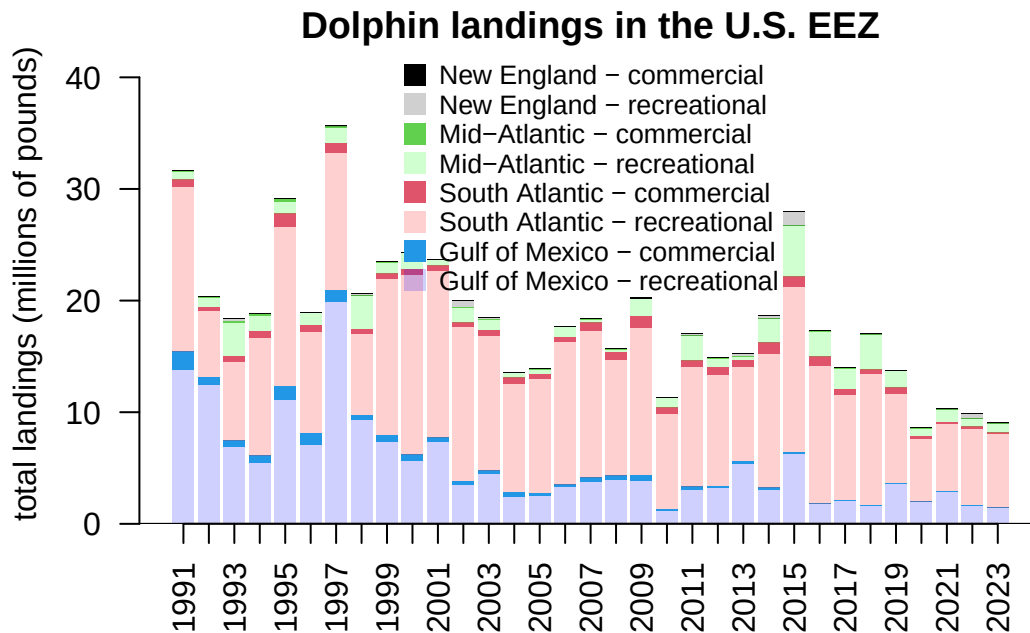
```
# select years -----
dat <- dat[which(rownames(dat) > 1990), ]

# define colors and plot -----
cols <- c("#0000FF30", 4, "#FF000030", 2, "#00FF0030", 3, "#00000030", 1)

par(mar = c(4, 4, 1, 0), mgp = c(2.5, 1, 0))
yrs <- as.numeric(rownames(dat))

b <- barplot(t(dat)/10^6, names.arg = yrs, col = cols, ylim = c(0, 43),
             border = NA, las = 2, ylab = "total landings (millions of pounds)",
             main = "Dolphin landings in the U.S. EEZ")
legend("top", c("New England - commercial", "New England - recreational",
               "Mid-Atlantic - commercial", "Mid-Atlantic - recreational",
               "South Atlantic - commercial", "South Atlantic - recreational",
               "Gulf of Mexico - commercial", "Gulf of Mexico - recreational"),
       col = cols[8:1], pch = 15, pt.cex = 1.5, bty = "n", y.intersp = 0.9, cex = 0.85)
axis(1, at = b, lab = rep("", length(b)), pos = 0)
abline(h=0)
```

4 NOAA Fisheries landings data



4.4 Updated Gulf/Atlantic parsing of data

There are some issues with the way that the FOSS system splits out Gulf and Atlantic catches by Monroe County, which is on the border between the two jurisdictions. We received data from Mike Larkin (SERO) which has the correct split out of data for Monroe County between the Gulf and Atlantic. We compare the total recreational catch for the Gulf and Atlantic between the FOSS data and the updated data. The numbers are very close; the average difference from the FOSS data is 2%.

```
ml <- read.table("data/RecreationalDolphinLandings_SAandGOM_MLarkin.csv", header = T, sep = ",")
ml <- ml[which(ml$Year > 1990), ]

# select same years for FOSS data set
dat <- dat[which(yrs <= max(ml$Year)), ]

# check that years match
summary(rownames(dat) == ml$Year)
```

Mode TRUE

4 NOAA Fisheries landings data

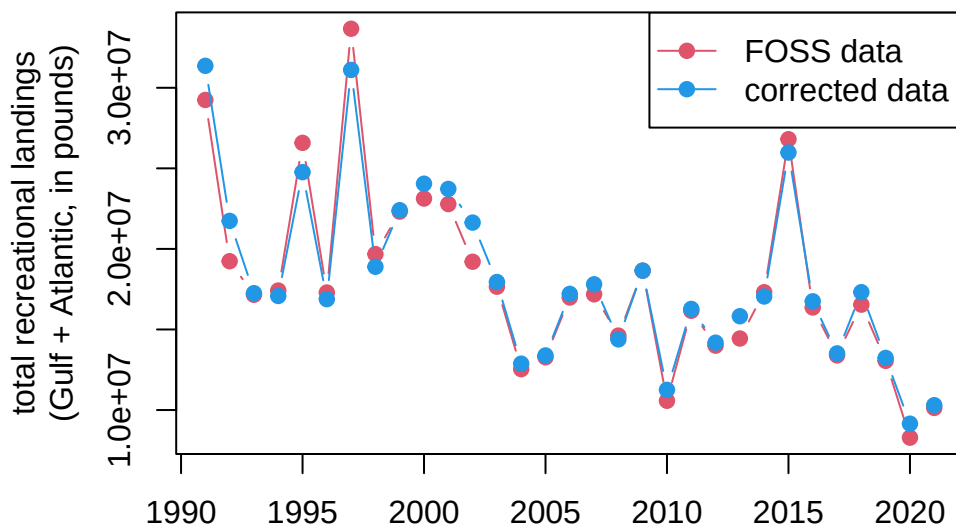
logical 31

```
yrs <- as.numeric(rownames(dat))

# check that totals match
old <- dat$rec_Gulf + dat$rec_SA + dat$rec_MA + dat$rec_NE
new <- ml$GOM + ml$ATL

# plot totals
par(mar = c(4, 6, 2, 1), mgp = c(2.5, 1, 0))
matplot(yrs, cbind(old, new), xlab = "", lty = 1, col = c(2, 4), type = "b", pch = 19,
        ylab = "total recreational landings\n(Gulf + Atlantic, in pounds)",
        main = "Comparison of FOSS data with data corrected for Monroe County")
legend("topright", c("FOSS data", "corrected data"), col = c(2, 4), pch = 19, lty = 1)
```

Comparison of FOSS data with data corrected for Monroe



```
# average % difference between FOSS and updated data
mean((old - new)/old*100)
```

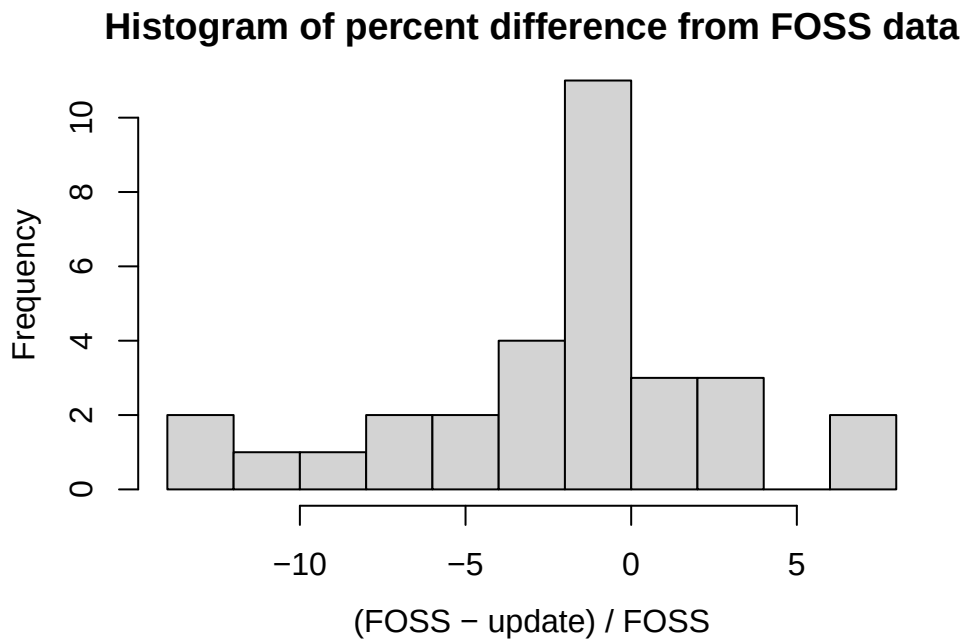
[1] -2.023886

4 NOAA Fisheries landings data

```
# min and max % difference between FOSS and updated data  
range((old - new)/old*100)
```

```
[1] -13.008552  7.587789
```

```
hist((old - new)/old*100, main = "Histogram of percent difference from FOSS data", breaks =  
      xlab = "(FOSS - update) / FOSS")
```



```
# replace FOSS data with update data  
datnew <- dat  
datnew$rec_Gulf <- ml$GOM  
datnew$rec_SA <- ml$ATL - (datnew$rec_MA + datnew$rec_NE)  
  
# old data set  
tail(dat)
```

	rec_Gulf	com_Gulf	rec_SA	com_SA	rec_MA	com_MA	rec_NE	com_NE
2016	1777348	37411	12357817	828133	2212513	61336	16499	15747
2017	2064106	65141	9457460	532973	1820541	16179	48607	30969

4 NOAA Fisheries landings data

2018	1626095	56188	11713866	435059	3104018	20514	97107	12667
2019	3567170	83956	7967978	563132	1518629	48224	66	16918
2020	1952241	26332	5653294	260311	679556	14040	0	1880
2021	2893817	26570	6061024	178822	1051785	25318	123038	11421

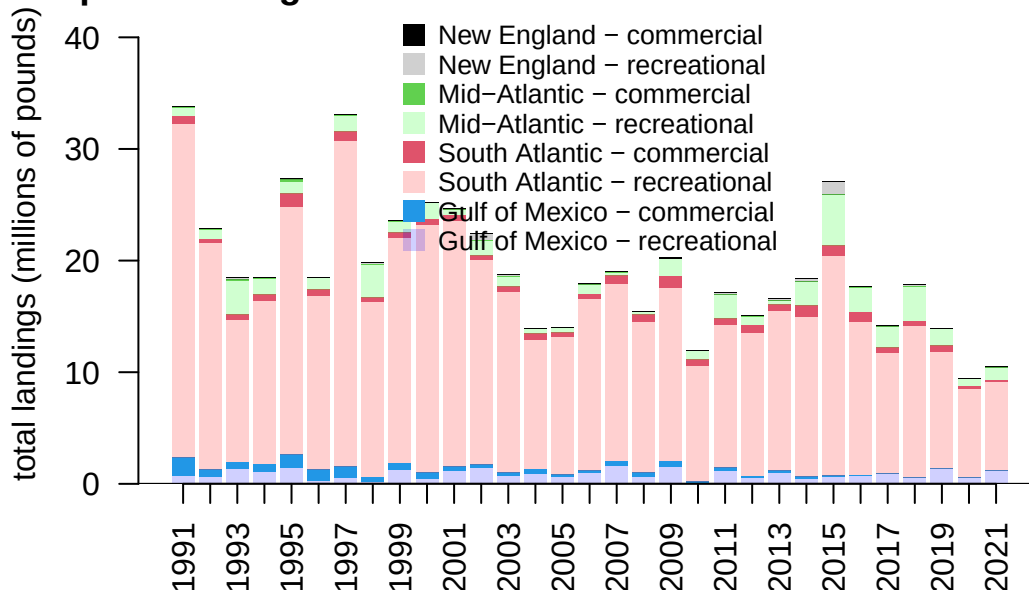
```
# updated data set  
tail(datnew)
```

	rec_Gulf	com_Gulf	rec_SA	com_SA	rec_MA	com_MA	rec_NE	com_NE
2016	751499	37411	13766851	828133	2212513	61336	16499	15747
2017	868489	65141	10777104	532973	1820541	16179	48607	30969
2018	508570	56188	13603875	435059	3104018	20514	97107	12667
2019	1304865	83956	10410599	563132	1518629	48224	66	16918
2020	555695	26332	7916891	260311	679556	14040	0	1880
2021	1174412	26570	7950716	178822	1051785	25318	123038	11421

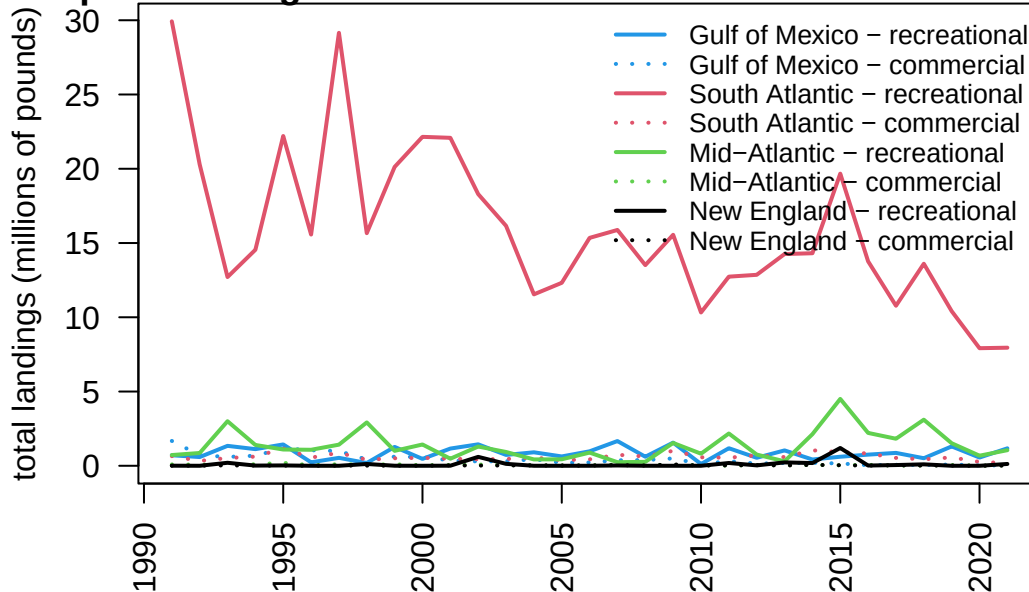
4.5 Plots with updated data

We replace the FOSS values for Gulf and South Atlantic recreational catches with these new values and re-plot the full data set. We can also plot the landings as a line graph so that the relative magnitudes of catch across region and sector are more easily compared.

Dolphin landings in the U.S. EEZ -- correct Monroe Coun



Dolphin landings in the U.S. EEZ -- correct Monroe Coun



5 SST

6 Analysis of ocean temperature trends

Coming soon